An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

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Abstract

Three-variable first-order logic (FO^3) and the calculus of relations (also known as relation algebra) define the same binary queries, an equivalence that goes back to Tarski's work in the 1940s. The classical translation from FO^3 into the calculus is single-exponential, but it has remained open whether this blowup is necessary. We prove that it is, even over finite structures and even when the translated terms may share subterms. Our proof introduces a new technique: we modify one finite structure in exponentially many small ways that each change the answer of a short FO^3 formula. In any equivalent term, only compositions of binary relations can notice these modifications, and each composition can notice at most two. The same technique gives an exponential succinctness gap between FO^3 and FO^2 sentences, and exponential lower bounds for matrix query languages, where the analogous question was also open.

1 Introduction

A classical result of Tarski from the 1940s [TG87, Mad21] states that three-variable first-order logic (**FO^3**) and the calculus of relations (**CoR**), a variable-free algebra of binary relations, express the same binary queries. The known translations between them have asymmetric size bounds:

$$\varphi(x,y) \in \mathrm{FO}^3 \xrightarrow[\textbf{optimal?}]{|t| = 2^{O(|\varphi|)}} t(x,y) \in \mathrm{CoR}, \qquad t(x,y) \in \mathrm{CoR} \xrightarrow[\textbf{optimal}]{|\varphi| = O(|t|)} \varphi(x,y) \in \mathrm{FO}^3.$$

Whether this exponential blowup is unavoidable has remained unknown [FGL$^+$15, Nak22, NU27]. **We resolve this question: the exponential growth is optimal**. Before stating our result precisely in §1.1, we review the importance of FO^3 and this translation in logic and computation.

Expressive power of FO^3. FO^3 is singular among the finite-variable fragments FO$^\ell$. FO^3 can express the transitivity law $\forall x \forall y \forall z\, (E(x,y) \wedge E(y,z) \to E(x,z))$, and thereby, together with irreflexivity and the absence of maximal elements, force infinite models. In contrast, FO^2 cannot express transitivity and has the finite model property [Mor75]. Over words, the variable hierarchy of FO collapses at three: every FO-definable property of words can be expressed in FO^3 [IK89], whereas FO^2 is strictly less expressive [EVW02]. Remarkably, three variables even suffice to formalize all of mathematics: Tarski and Givant showed that set theories such as ZFC can be formalized by CoR equations [TG87, Giv06, Mad21]. This expressive strength is also reflected in computational complexity. For FO^3, satisfiability and finite satisfiability are undecidable even over a single binary relation [KMW62, Nak19, Ryb22]; for FO^2, both are decidable [Mor75]. For a fixed sentence, model counting can be **#P$_1$-complete** in FO^3 [BVGS15] but takes polynomial time in FO^2 [VMD14]. Asser's longstanding question of whether first-order spectra (sets of finite-model sizes) are closed under complement [Ass55] is equivalent to the open problem **NE** = **coNE** [JS74]. Resolving it for FO^3 over a single binary relation would already settle the general problem [KT15, KT18].

Algorithmic power of the FO^3*-to-*CoR *translation.* The preceding results illustrate the logical and computational power of FO^3. Its exceptional status persists even in model checking, where every fixed FO sentence can be evaluated in polynomial time. A sentence $\varphi \in \mathrm{FO}^\ell$ can be evaluated on an n-element structure in $O(|\varphi| n^\ell)$ time [Var95]. For $\ell \geq 4$, this exponent is conditionally optimal under the hyperclique hypothesis [NU27, Corollary 32]. For FO^3, however, the cubic barrier can be broken: FO^3 model checking takes $2^{O(|\varphi|)} n^\omega$ time [NU27, Theorem 10], where $\omega < 2.3714$ is the exponent of matrix multiplication [ADV$^+$25]. **The key to this speedup is the classical FO^3-to-CoR translation**. It transforms a formula φ into a CoR term t, which can be evaluated in $O(|t| n^\omega)$ time because a composition is a Boolean matrix product and every other operation takes $O(n^2)$ time. Thus, the translation size directly affects the running time. If every FO^3 formula admitted a polynomial-size CoR representation, every such query would have a compact algebraic evaluation plan, executable in $\mathrm{poly}(|\varphi|) n^\omega$ time once compiled. Our result rules out this possibility: some FO^3 queries inherently require exponentially large CoR representations, even with shared subterms.

1.1 Main results

Setup: reviewing CoR**.** We first recall the operations of CoR. Over a nonempty domain $\mathbb{D}$, **CoR terms** denote binary relations and are built from input relation symbols, the constants $\mathbf{0} \coloneqq \varnothing$, $\mathbf{1} \coloneqq \mathbb{D}^2$, and $\mathbf{id} \coloneqq \{(u,u) : u \in \mathbb{D}\}$, using union $X \cup Y$, intersection $X \cap Y$, complement $\overline{X} \coloneqq \mathbb{D}^2 \setminus X$

(also use $\neg X$ as $\overline{X}$), converse $X^\smile := \{(v,u) : (u,v) \in X\}$, and composition $X;Y$, defined by

$$X;Y := \{(u,v) : \text{some } w \in \mathbb{D} \text{ satisfies } (u,w) \in X \text{ and } (w,v) \in Y\}.$$

Composition implements the elimination step $\exists z\,(P(x,z) \wedge Q(z,y)) \Leftrightarrow (P;Q)(x,y)$; a vertex z with $P(x,z)$ and $Q(z,y)$ is a **witness** for (x,y).

A CoR term T is a syntax tree with $|T|$ nodes and $\#_;(T)$ compositions. A **relational circuit** C is a DAG (rather than a tree) and may share subterms, and $|C|$ and $g_;(C)$ count its distinct nodes and composition gates. For an FO formula φ, we write $T \equiv_{\text{fin}} \varphi$ or $C \equiv_{\text{fin}} \varphi$ for equivalence on all finite structures over the input signature, identifying formulas and relational expressions with the binary queries they define.

Our hard family and its translation into CoR **(upper bound).** We use the following single-quantifier positive FO^3 formula over $2k$ binary relation symbols $P_1, \dots, P_k, Q_1, \dots, Q_k$:

$$G_k(x,y) := \exists z \bigwedge_{i=1}^{k} \Big[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z))\Big].$$

The signature of G_k consists only of binary relations, but grows with k. It asks whether the queried pair (x,y), the **read edge**, **accepts** some **witness edge** (x,z): at every coordinate i, one of the two edges satisfies P_i and the other Q_i. The choice may differ across coordinates, but the same witness z must serve all of them.

The natural way to translate G_k into CoR is to expand it into disjunctive normal form (DNF), choosing one of the two disjuncts at each coordinate. Let $[k] := \{1, \dots, k\}$. For $A \subseteq [k]$, let $A^c := [k] \setminus A$ and $H_A := (\bigcap_{i \in A} P_i) \cap (\bigcap_{i \notin A} Q_i)$, where an empty intersection is $\mathbf{1}$.

$$\begin{aligned} G_k(x,y) &\equiv \exists z \bigvee_{A \subseteq [k]} \Big[\bigwedge_{i \in A} (P_i(x,y) \wedge Q_i(x,z)) \wedge \bigwedge_{i \notin A} (Q_i(x,y) \wedge P_i(x,z))\Big] \equiv \exists z \bigvee_{A \subseteq [k]} [H_A(x,y) \wedge H_{A^c}(x,z)] \\ &\equiv \bigvee_{A \subseteq [k]} [H_A(x,y) \wedge \exists z\, H_{A^c}(x,z)] \equiv \Big(\bigcup_{A \subseteq [k]} [H_A \cap (H_{A^c};\mathbf{1})]\Big)(x,y). \end{aligned}$$

The last step uses row projection: $(X;\mathbf{1})(x,y) \Leftrightarrow \exists z\, X(x,z)$. The DNF yields a CoR term T_k with 2^k compositions and $O(k2^k)$ nodes. By sharing partial intersections among the H_A, it also yields a relational circuit C_k with 2^k composition gates and only $O(2^k)$ nodes. Thus, we have $\#_;(T_k) = g_;(C_k) = 2^k$, $|T_k| = O(k2^k)$, and $|C_k| = O(2^k)$.

The above translation illustrates the essence of the classical single-exponential FO^3-to-CoR translation (see also Proposition D.1).

Main result: lower bound. One might hope that the DNF expansion can be substantially compressed using more sophisticated CoR terms. Surprisingly, the straightforward translation is already optimal within a factor of two in the number of compositions.

Theorem 1.1 (Exponential lower bounds for CoR)**.** *For every $k \geq 1$, every relational circuit $C \equiv_{\text{fin}} G_k$ has at least 2^{k-1} composition gates, and every* CoR *term $T \equiv_{\text{fin}} G_k$ has at least 2^{k-1} compositions.*

It counts only compositions: all other operations are unrestricted and free. Since G_k has only $8k$ syntax-tree nodes, the theorem establishes an unconditional exponential succinctness gap from FO^3 to CoR, even when the target is a circuit rather than a term. Together with the classical single-exponential upper bound, this settles the worst-case size of the FO^3-to-CoR translation.

Beyond the number of compositions, we determine the minimum size of equivalent CoR terms up to an asymptotically vanishing relative error: every CoR term $T \equiv_{\mathrm{fin}} G_k$ has at least $2k2^k - 1$ nodes, and factoring the conditions on the read edge into a binary decision tree gives one with $(2k+6)2^k - 5$ nodes (§5.2). The extra factor k comes from input occurrences that a term must repeat but a circuit can share: the circuit C_k above has only $O(2^k)$ nodes, so the optimal circuit size is $\Theta(2^k)$, whereas the optimal term size is $(2 + o(1))k2^k$.

Where the blowup begins. How large must the domain be before the exponential lower bound becomes unavoidable? We show that exponentially large domains already suffice, even for circuits specialized to a fixed domain size.

Proposition 1.2 (Domain size)**.** *For some absolute constant $C > 0$, for every $k \geq 1$ and $n \geq Ck4^k$, every relational circuit computing G_k on all n-element structures has at least 2^{k-1} composition gates.*

Whether the domain-size bound $Ck4^k$ is asymptotically optimal remains open. On the other hand, the full lower bound fails for $n < 2^k/(8k)$ (Proposition 4.3), leaving a gap between the known bounds.

1.2 Natural approaches that fail

The **fooling-set argument** [Yao79, LS81, KN97, RY20] is a standard technique for proving lower bounds in communication complexity. For a Boolean function $f\colon A \times B \to \{0,1\}$, a **fooling set** is a set $F \subseteq f^{-1}(1)$ such that any two distinct $(a,b),(a',b') \in F$ have a crossed input (a,b') or (a',b) of value 0. A **1-rectangle** is a product $A' \times B' \subseteq f^{-1}(1)$. Since a rectangle containing two points also contains their crossings, each 1-rectangle covers at most one point of F, so at least $|F|$ are needed.

One might try the same argument for our FO^3-to-CoR lower bound. Indeed, the earlier exponential lower bound for $\mathbf{\Sigma_1 CoR}$, even with shared subterms [Nak22, Theorems 6.1 and 6.12], can be viewed as a **relational fooling-set argument**. Although the published proof uses formula-size games in the style of Adler and Immerman [AI03], the underlying obstruction has a simple interpretation. Here Σ_1CoR allows complement only on input relations and **id** (after pushing it to the leaves): $\overline{R};\mathbf{1}$ is permitted, but $\overline{R;\mathbf{1}}$ is not.

Consider Nakamura's query of size $O(k)$, $\varphi_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k}(R_i(x,z) \vee R_i(z,y))$ [Nak22]. Let $I, J \subseteq [k]$ record the relation labels on the two witness edges (x,z) and (z,y). The witness condition is precisely $f_k(I,J) = 1 \iff I \cup J = [k]$, with the fooling set $F_k = \{(I, I^c) : \varnothing \subsetneq I \subsetneq [k]\}$ of size $2^k - 2$ for $k \geq 2$. Each point (I, I^c) is realized in [Nak22, §5] by a positive pointed structure $(\mathcal{S}_I; u_I, v_I)$ with a witness path $u_I \xrightarrow{I} w_I \xrightarrow{I^c} v_I$. There is a subtlety: these paths belong to different structures, so we cannot directly cross (u_I, v_I) and (u_J, v_J). The construction provides a common negative pointed structure $(\mathcal{N}; u^*, v^*)$, with $\mathcal{S}_I \models \varphi_k(u_I, v_I)$ for every I but $\mathcal{N} \not\models \varphi_k(u^*, v^*)$. Whenever $f_k(I, J^c) = 0$, the left witness half (u_I, w_I) and the right half (w_J, v_J) embed into $\mathcal{N}$ at $(u^*, c_{I,J})$ and $(c_{I,J}, v^*)$, respectively, using a vertex $c_{I,J}$.

How can we cross witnesses from different structures? The key is **preservation under embeddings**: every Σ_1CoR term V and embedding $\iota\colon \mathcal{A} \hookrightarrow \mathcal{B}$ satisfy

$$\iota([\![V]\!]_{\mathcal{A}}) \subseteq [\![V]\!]_{\mathcal{B}}, \quad \text{or equivalently,} \quad \mathcal{A} \models V(u,v) \implies \mathcal{B} \models V(\iota(u), \iota(v)). \tag{$\diamond$}$$

Now let T be a Σ_1CoR term equivalent to φ_k. A short structural induction on T (after pushing converses to the leaves, and using the embeddings to exclude endpoint witnesses) shows that, for each I, some composition subterm $X;Y$ of T satisfies (1) $\mathcal{S}_I \models X(u_I, w_I) \wedge Y(w_I, v_I)$; and (2) $\mathcal{N} \not\models (X;Y)(u^*, v^*)$. Suppose $\mathcal{S}_J \models X(u_J, w_J) \wedge Y(w_J, v_J)$ also holds for $I \neq J$. Since $(I, I^c), (J, J^c) \in F_k$, at least one of their crossed inputs has value 0. Suppose $f_k(I, J^c) = 0$; the other case is symmetric. The crossing embeddings and ($\diamond$) then give $\mathcal{N} \models (X;Y)(u^*, v^*)$ because $\mathcal{N} \models X(u^*, c_{I,J}) \wedge Y(c_{I,J}, v^*)$. This contradicts (2).

Thus, **such composition $X;Y$ plays the role of a 1-rectangle**, accounting for at most one fooling-set point. The argument extends to shared subterms [Nak22, §6.3].

Why complement defeats the argument. The crossing argument relies crucially on ($\diamond$), which fails under unrestricted complement. Indeed, $V := \overline{R;\mathbf{1}}$ violates ($\diamond$). $V(x,y)$ means $\neg\exists z.R(x,z)$ in FO^3. Let $\mathcal{A}$ have no R-successor of u, and obtain an extension $\mathcal{B}$ by adding a fresh R-successor. Then, $\mathcal{A} \models V(u,v)$ and $\mathcal{B} \not\models V(u,v)$.

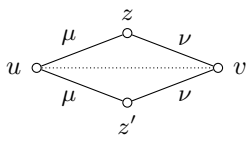
Furthermore, the above $2^k - 2$ positive instances $(\mathcal{S}_I; u_I, v_I)$ and the negative instance $(\mathcal{N}; u^*, v^*)$ can be distinguished by a single full-CoR term with only two compositions. Writing $E := \bigcup_i (R_i \cup R_i^\smile)$, consider $U := \neg((E \cap (E; \neg\mathbf{id})); \mathbf{1})$ where $\neg T$ means $\overline{T}$. $U(x,y)$ means $\neg[\exists y \exists z.(E(x,y) \wedge E(x,z) \wedge y \neq z)]$ in FO^3. This holds in every $\mathcal{S}_I$ but fails in $\mathcal{N}$, where u^* has at least two neighbors. Although U does not define φ_k, the old test instances themselves cannot certify an exponential lower bound for full CoR.

1.3 Technical overview

Rather than extending the fooling-set argument in §1.2, we abandon witness crossing and embeddings altogether. Instead, we compare a reference structure with its local modifications using exact predictions that commute with complement.

Our technique: The two-witness property. We use a complete undirected graph $\mathbb{G} = (V, \chi)$ whose edges are colored by the 2^k subsets of $[k]$. We write $\chi(u,v) = \chi(v,u)$ for the color of the edge between distinct vertices u and v, and use λ, μ, ν for colors. The key property of $\mathbb{G}$ is the **two-witness property**: for all distinct $u, v \in V$ and all colors μ, ν,

at least two vertices $z \notin \{u, v\}$ satisfy $\chi(u,z) = \mu$ and $\chi(z,v) = \nu$.



We call such vertices z **witnesses**. A coloring with this property exists whenever $|V| \geq n_0(2^k)$, where $n_0(N) := \lceil 12N^2 \ln(2N) \rceil + 3$, since a uniformly random coloring has it with positive probability (Lemma 3.1). These colorings go back to Jipsen, Maddux, and Tuza [JMT95].

Roadmap. In §1.3.1, we construct 2^k folds of the reference structure $\mathcal{M}_0$, each changing the answer of G_k at an unchanged edge. In §1.3.2, we show that every circuit gate on $\mathcal{M}_0$ is constant

on each edge-color class and on the diagonal. This allows us to define predictions under each fold. Boolean operations, including complement, and converse preserve these predictions; only composition can violate them. In §1.3.3, we show that, thanks to the two-witness property, each composition can violate the prediction for at most two folds. Since every fold must be detected, any circuit for G_k requires at least 2^{k-1} compositions, even with sharing.

1.3.1 Folding the reference structure

We use the reference coloring χ to define an input structure $\mathcal{M}_0$ for G_k. On an edge of color μ, we interpret P_i as true exactly when $i \in \mu$, and Q_i exactly when $i \notin \mu$. No input relation holds on loops, which carry the separate label **id**. Consequently, a read edge of color μ accepts exactly witness edges of color μ^c: for distinct u, v,

$$\mathcal{M}_0 \models G_k(u,v) \iff \exists z \notin \{u,v\}.\ \chi(u,z) = \chi(u,v)^c.$$

By the two-witness property, every vertex has edges of every color, so G_k holds at every non-loop pair of $\mathcal{M}_0$.

Fix a vertex $\hbar \in V$, the **hub**, and for each color μ choose v_μ with $\chi(\hbar, v_\mu) = \mu$. For each color λ, let χ_λ be obtained from χ by recoloring every λ-colored edge incident to $\hbar$ as λ^c, leaving all other edges unchanged. Let $\mathcal{M}_\lambda$ be the input structure induced by χ_λ in the same way as $\mathcal{M}_0$. In $\mathcal{M}_\lambda$, the hub has no edge of color λ, but the read edge $(\hbar, v_{\lambda^c})$ remains unchanged (Figure 1).

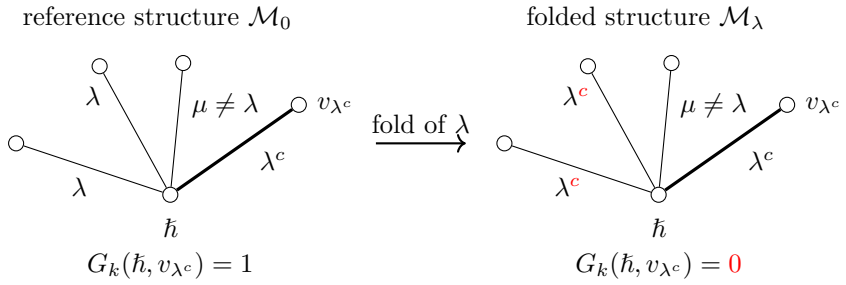


Figure 1: The fold of λ. Only edges at the hub $\hbar$ are drawn.

The read edge $(\hbar, v_{\lambda^c})$ requires a λ-colored edge at $\hbar$; the fold removes all such edges. Hence

$$\chi_\lambda(\hbar, v_{\lambda^c}) = \chi(\hbar, v_{\lambda^c}) = \lambda^c,\ \mathcal{M}_0 \models G_k(\hbar, v_{\lambda^c}),\ \mathcal{M}_\lambda \not\models G_k(\hbar, v_{\lambda^c}).$$

We call $(\mathcal{M}_0, \mathcal{M}_\lambda, (\hbar, v_{\lambda^c}))$ the **test** for λ. Thus G_k yields 2^k tests, one for each color.

1.3.2 Circuits see only colors

Color-determined relations. A relation $X \subseteq V^2$ is **color-determined** if $\chi(u,v) = \chi(u',v') \implies X(u,v) = X(u',v')$ holds. Here loops carry the separate label **id**. Equivalently, X is a union of edge-color classes, possibly together with the diagonal. Write $X[\mu]$ for its common value on edges of color μ, $X[\mathbf{id}]$ for its value on loops.

Every gate of an arbitrary CoR circuit computes a color-determined relation on $\mathcal{M}_0$ (Lemma 3.2). We prove this by induction on the circuit gates. Input relations and constants are color-determined, while Boolean operations and converse preserve this property. The only nontrivial step is composition $X;Y$.

For distinct u, v, the two-witness property realizes every pair of colors μ, ν along a path $u \xrightarrow{\mu} z \xrightarrow{\nu} v$ with $z \notin \{u, v\}$. The case $z \in \{u, v\}$ depends only on $\chi(u, v)$ and the diagonal values of X, Y. Hence $(X;Y)(u, v)$ depends only on $\chi(u, v)$. For a loop (u, u), both edges of a path $u - z - u$ have the same color, and every color occurs at every vertex. Together with the case $z = u$, this makes $(X;Y)(u, u)$ independent of u, completing the induction.

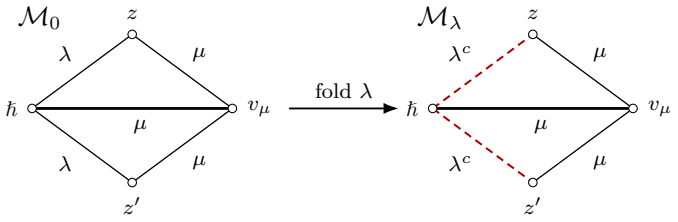
Predicting and noticing a fold. A color-determined relation X assigns a value $X[\mu]$ to each color μ. To predict the effect of a fold, we keep these values fixed and read them at the new edge colors:

$$\textbf{(Prediction)} \quad X^{(\lambda)}(u, v) := X[\chi_\lambda(u, v)].$$

Thus a recolored edge reads $X[\lambda^c]$ rather than $X[\lambda]$; all other pairs retain their values. This predicts what a gate would compute if its color-dependent behavior on $\mathcal{M}_0$ remained unchanged after the fold; its actual value on $\mathcal{M}_\lambda$ may differ.

Input relations and constants behave exactly as predicted. Moreover, Boolean operations and converse commute with prediction (Proposition 3.3); in particular, $\overline{X^{(\lambda)}} = (\overline{X})^{(\lambda)}$. Unlike the embedding inclusion ($\diamond$) of §1.2, this equality survives unrestricted complement.

Figure 2 compares three operations for any two distinct colors λ, μ: X selects the λ-colored edges, and Y the μ-colored edges (both exclude loops). The thick edge $e = (\hbar, v_\mu)$ is unchanged by the fold of λ, which removes all $X^{(\lambda)}$-edges at $\hbar$.



Operation	Predicted	Actual	Notice?
$X;Y$	1	0	Yes
$X \cup Y$	1	1	No
$\overline{X}$	1	1	No

Figure 2: At $e = (\hbar, v_\mu)$, only composition notices the fold of λ. Only two witness paths are shown; all λ-edges at $\hbar$ are recolored to λ^c.

An operation **notices** a fold at given color-determined operands if applying it to their predictions differs from predicting its result. In Figure 2, composition notices the fold at e, since

$$(X^{(\lambda)}; Y^{(\lambda)})(e) = 0 \neq 1 = (X;Y)^{(\lambda)}(e).$$

By contrast, union and complement do not notice the fold. More generally, no Boolean operation or converse can notice any fold (Proposition 3.3).

Every fold must be noticed by some composition gate. Now let C be any circuit computing G_k. If no composition gate notices the fold of λ, Lemma 3.4 implies that the output on $\mathcal{M}_\lambda$ is the prediction of the output on $\mathcal{M}_0$. Since the test edge $e_\lambda = (\hbar, v_{\lambda^c})$ is unchanged,

$$[\![C]\!]_{\mathcal{M}_\lambda}(e_\lambda) = \left([\![C]\!]_{\mathcal{M}_0}\right)^{(\lambda)}(e_\lambda) = [\![C]\!]_{\mathcal{M}_0}(e_\lambda),$$

contradicting the test for λ, whose outputs are 0 on $\mathcal{M}_\lambda$ and 1 on $\mathcal{M}_0$.

1.3.3 Each composition notices at most two folds

We now show that any composition $X;Y$ of color-determined relations notices at most two folds (Lemma 3.5). The two-witness property first rules out noticing at pairs away from the hub.

Away from the hub (Figure 3). At pairs (u, v) with $u, v \neq \hbar$, composition agrees with prediction. Figure 3 uses the same coloring, operands X, Y, and fold as Figure 2, but observes $e' = (u, v_\mu)$.

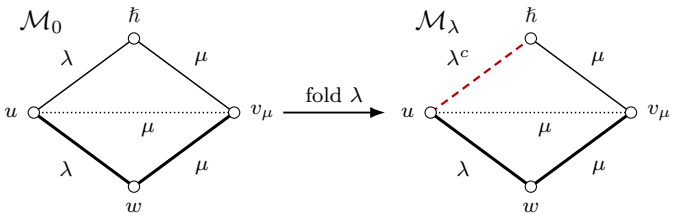


Figure 3: Away from the hub, the fold invalidates the path through $\hbar$ but leaves the second witness path through w unchanged. Only these two witness paths are shown.

In $\mathcal{M}_0$, both paths have colors (λ, μ). After the fold, the path through $\hbar$ is invalidated but the path through w survives. Hence: $(X^{(\lambda)}; Y^{(\lambda)})(e') = 1 = (X;Y)^{(\lambda)}(e')$.

More generally, between any distinct $u, v \neq \hbar$, the two-witness property preserves every two-color path pattern via a witness other than $\hbar$. Loops away from $\hbar$ retain edges of every color, and endpoint witnesses are unchanged. Thus composition cannot violate prediction away from the hub.

At the hub (Figure 2 revisited). Only pairs involving $\hbar$ remain. In Figure 2, X holds only on λ and Y only on μ, where $\lambda \neq \mu$. Their composition notices the folds of λ and μ at $(\hbar, v_\mu)$ and $(v_\lambda, \hbar)$, respectively.

Why can it not notice a third fold? To notice λ at $(\hbar, v)$, X must hold on λ and no other off-diagonal color. Indeed, if $X[\nu] = 1$ for some $\nu \neq \lambda$ and $Y[\tau] = 1$ for some color τ, the two-witness property supplies a path $\hbar \xrightarrow{\nu} z \xrightarrow{\tau} v$ that survives the fold of λ. Thus z remains a witness for $(X^{(\lambda)}; Y^{(\lambda)})(\hbar, v)$. Likewise, noticing a distinct fold μ at $(u, \hbar)$ requires Y to hold only on μ. Hence noticing two distinct folds forces X and Y to be supported on $\{\lambda\}$ and $\{\mu\}$, respectively. No third color occurs in either support, and their supports are disjoint, ruling out an additional fold noticed at $(\hbar, \hbar)$. If X and Y share a color instead, all possible noticed folds must coincide. Thus $X;Y$ notices at most two folds.

Summary. Every one of the 2^k folds requires a noticing composition gate (§1.3.2), whereas each distinct composition gate notices at most two (Lemma 3.5). Hence $2^k \leq 2g_;(C)$, proving the lower bound of Theorem 1.1 even with sharing.

1.4 Further results

All lower bounds in Table 1 follow the scheme of §1.3; the entries marked exact are matched by the DNF and by Proposition B.1. Only two ingredients change. The bound two of Lemma 3.5 is replaced by the number of folds that an operation combining values over a witness can notice, while

every other operation notices none; and a query yields its tests as in §1.3. The term-size bounds add the occurrence count of §5.2.

Target or setting	Folds noticed per gate	Lower bound for G_k
Relational circuits	composition 2, projection 1	$g_{;}(C) \geq 2^{k-1}$ (Thm. 1.1)
Without converse	composition 1 (directed input)	$g_{;}(C) \geq 2^k$, exact (Cor. 4.2)
CoR equations for $G_k^\forall$	as for circuits	$2^{k-1}-1$ compositions (Cor. 1.3)
FO^2 formulas, sentences	quantifier 1	2^k quantifiers (§5, Thm. 1.4)
Matrix languages, breadth β	matrix–matrix 3β, matrix–vector β	$\beta(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)) \geq 2^k$ (Thm. 1.6)
With closure X^*	closure 1	$2g_{;}(C) + g_*(C) \geq 2^k$ (Prop. 1.7)
Randomized, error ε	as for circuits	factor $1-2\varepsilon$ (Cor. 1.8)
Random inputs	as for circuits	success at most $\frac{1}{2} + g/2^k + o(1)$, exact (Cor. 1.9)

Table 1: One argument, many targets. The middle column bounds the number of folds that a single operation of each kind can notice; all other operations notice none. Every bound holds at each domain size at which the reference structure exists.

Sentences and equations. Our bound also applies to Tarski's correspondence in its equational form, which matches three-variable sentences with equations between relation terms; in this form he formalized set theory [TG87]. A sentence has a single value on each structure, and our tests compare structures: $\mathcal{M}_0$ satisfies G_k at every pair of distinct vertices, while each $\mathcal{M}_\lambda$ has a failing pair, so every test changes the value of the sentence

$$G_k^\forall \coloneqq \forall x\, \forall y\, (x = y \vee G_k(x, y)).$$

An equation $T = S$ holds exactly when $\overline{\mathbf{1}; ((T \cap \overline{S}) \cup (\overline{T} \cap S)); \mathbf{1}}$ is full, which costs two one-sided compositions, so the argument of Theorem 1.1 applies with a loss of one.

Corollary 1.3 (CoR equations)**.** *If a* CoR *equation $T = S$ holds on exactly those finite structures that satisfy $G_k^\forall$, then every relational circuit with gates computing T and S has at least $2^{k-1}-1$ composition gates.*

The DNF gives the equation $\mathbf{id} \cup \bigcup_{A \subseteq [k]} [H_A \cap (H_{A^c}; \mathbf{1})] = \mathbf{1}$ with 2^k compositions, so this bound is tight up to a factor of two.

Three versus two variables. To our knowledge, the succinctness of three versus two variables had been studied only on linear orders, where three-variable sentences translate into two-variable sentences of polynomial size [GS05]. Exponential gaps were known for more variables: four versus three on linear orders [GS05], and $\ell+1$ versus ℓ for every $\ell \geq 3$ on general structures with higher-arity relations [VS24]. We show that on general finite structures, three variables are exponentially more succinct than two, already for sentences over binary relations. In an FO^2 formula, every subformula defines a binary relation, like a gate of a relational circuit: connectives notice nothing, and a quantifier is a projection, $\exists y\, \psi \equiv \psi; \mathbf{1}$ or $\exists x\, \psi \equiv \mathbf{1}; \psi$, which notices at most one fold because one operand is constant (§1.3.3). Hence an FO^2 sentence equivalent to $G_k^\forall$ needs a separate quantifier for each of the 2^k tests.

Theorem 1.4 (Three-variable versus two-variable sentences)**.** *For each $k \geq 1$, the sentence $G_k^{\forall}$ has size $O(k)$ and quantifier depth three over $2k$ binary relations. It is* FO^2*-definable, but every equivalent* FO^2 *sentence over finite structures has at least 2^k quantifiers and $k2^k$ relation-atom occurrences. Its minimum* FO^2 *tree size is $\Theta(k2^k)$.*

The DNF of §1.1 defines G_k in FO^2: writing $H_A(x, y)$ for the conjunction defining H_A gives

$$G_k(x, y) \equiv \bigvee_{A \subseteq [k]} [H_A(x, y) \wedge \exists y\, H_{A^c}(x, y)].$$

This expansion has 2^k quantifiers, and substituting it into $G_k^{\forall}$ gives $2^k + 2$, within two of the lower bound. The first conjunct checks the read edge. In the second, the endpoint y is no longer needed, so its name can be reused for the witness; the binding does not extend to the first conjunct. The short FO^3 formula keeps the read edge and the witness edge in scope together, and so can leave the order at each coordinate open without this expansion. For the binary query G_k, the same tests show that the expansion is optimal.

CoR versus two variables. Our method also separates CoR from FO^2, on a structure with fewer witnesses. On the reference structure, a composition notices at most two folds, so the tests there bound CoR and FO^2 alike, up to a factor of two. A projection, however, still notices at most one fold when every vertex has at least two edges of each atomic type, whereas a composition may then notice many. Let T_k be a CoR term of size $O(k)$ that relates x to y when some R-successor z of x satisfies $P_i(z, z) \leftrightarrow P_i(y, y)$ for every $i \in [k]$. If the fold of $A \subseteq [k]$ removes the R-edges from a hub to the vertices v with $\{i : P_i(v, v)\} = A$, the composition with witness z notices all 2^k folds (§5.4).

Proposition 1.5 (CoR versus FO^2)**.** *For every $k \geq 1$, the* CoR *equation* $\mathbf{id} \cup T_k = \mathbf{1}$ *is* FO^2*-definable, but every equivalent* FO^2 *sentence over finite structures has at least 2^k quantifiers. Every* FO^2 *formula defining T_k also has at least 2^k quantifiers, and this bound is attained.*

Together with Theorems 1.1 and 1.4 and Corollary 1.3, Proposition 1.5 separates FO^3, CoR, and FO^2 exponentially, for binary queries and for sentences, in every direction not covered by the linear translations from FO^2 to CoR [Nak22] and from CoR to FO^3.

Arity elimination in matrix query languages. Brijder, Gyssens, and Van den Bussche asked whether eliminating three-attribute intermediate relations in favor of matrix expressions necessarily incurs an exponential increase in size [BGV19, p. 9]. Matrix query languages replace composition by **aggregation**

$$(X \cdot Y)(u, v) \coloneqq \bigvee_w \kappa(X(u, w), Y(w, v)),$$

where $\bigvee$ is the join of an idempotent semiring $(S, \vee, \otimes, 0_S, 1_S)$ and the **kernel** κ is an arbitrary function of two scalars; with $\kappa = \otimes$, aggregation of matrices and vectors is the usual matrix product. The semiring has **join breadth** at most β if every finite join equals the join of at most β of its summands. We prove an exponential increase over every nontrivial commutative idempotent semiring of finite join breadth β. Write $\mathcal{G}_k$ for G_k with disjunction, conjunction, and existential quantification replaced by $\vee$, $\otimes$, and the join over witnesses. It has a positive ARA(3) expression of size $O(k)$, where ARA(3) denotes relational algebra on relations whose tuples are annotated

with semiring values, with at most three attributes per intermediate relation. A typed matrix circuit C computes with $n \times n$ matrices, vectors, and scalars over the semiring; let $\#_{\mathrm{mv}}(C)$ count its matrix–vector and vector–matrix aggregation gates, and $\#_{\mathrm{mm}}(C)$ its matrix–matrix aggregation gates.

Theorem 1.6 (Exponential arity-elimination overhead)**.** *For every such semiring and $k \geq 1$, every typed matrix circuit C computing $\mathcal{G}_k$ on all finite inputs satisfies*

$$\beta(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)) \geq 2^k,$$

and so has at least $2^k/(3\beta)$ aggregation gates. This holds with arbitrary entrywise functions, each applied in the same way at every entry, and arbitrary kernels in the model of §6. Conversely, some typed matrix circuit with 2^k matrix–vector products computes $\mathcal{G}_k$ on all inputs.

Circuits have fixed wiring and no vertex-dependent constant arrays; a kernel receives only its two scalar operands. Transpose and diagonal operations are also allowed. For max–min and tropical semirings, $\beta = 1$, so the 2^k matrix–vector products of the upper bound are optimal among circuits without matrix–matrix products. In the proof, at each of the three kinds of pairs at the hub in §1.3.3, $(\hbar, v)$, $(u, \hbar)$, and $(\hbar, \hbar)$, the existence of an outside witness becomes a join over the witnesses. Each join equals the join of at most β of its summands, and a fold can change the join only by removing one of these summands, so at most β folds change each join. Thus a matrix–matrix aggregation notices at most 3β folds, and a matrix–vector or vector–matrix aggregation at most β, whatever the kernel. For nonidempotent semirings such as $(\mathbb{N}, +, \times)$, where join breadth is not available, §6.4 proves $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C) \geq 2^k$ for the analogue of $\mathcal{G}_k$ with sums in place of joins, against circuits restricted to positive operations.

Reflexive–transitive closure. The reflexive–transitive closure $X^* \coloneqq \bigcup_{m \geq 0} X^m$ of an $n \times n$ Boolean matrix can be computed in the time of one matrix product, $O(n^\omega)$ [FM71, Mun71, Fur70], although it can replace up to $n - 1$ successive compositions and expresses reachability, which is not first-order. It is thus a natural primitive to add when products are the cost, but it does not avoid the blowup.

Proposition 1.7 (Closure does not help)**.** *Let C be a relational circuit that may also use $X \mapsto X^*$, and let $g_*(C)$ count its distinct closure gates. If $C \equiv_{\mathrm{fin}} G_k$, then $2g_;(C) + g_*(C) \geq 2^k$; in particular, at least 2^{k-1} gates of C are compositions or closures.*

Closure therefore does not rescue the speed-up of fast matrix multiplication: once $n \geq n_0(2^k)$, the query G_k of size $O(k)$ still needs 2^{k-1} operations that each cost a matrix product.

Fixed domain sizes, bounded error, and random inputs. The reference structure exists separately at each $n \geq n_0(2^k)$, and the folds keep its domain, so the bounds hold for **size-specific** circuits, chosen separately for each domain size, as in Proposition 1.2. A randomized circuit is an input-independent distribution of circuits. Its error is **pointwise**: for each input and each pair of vertices, the probability of a wrong answer is at most ε.

Corollary 1.8 (Fixed sizes and bounded error)**.** *Fix $k \geq 1$, $n \geq n_0(2^k)$, and $0 \leq \varepsilon < 1/2$. Every input-independent distribution of relational circuits computing G_k on all n-element inputs with pointwise error at most ε satisfies*

$$\mathbb{E}\, g_{;}(C) \geq (1 - 2\varepsilon)2^{k-1}.$$

In particular, each exact circuit at each such size needs at least 2^{k-1} composition gates, even when chosen specifically for n.

The size threshold is $O(k4^k)$, and by Proposition 4.3 it cannot be lowered below $2^k/(8k)$. The factor $1 - 2\varepsilon$ comes from comparing the two structures of a test: a circuit that cannot distinguish them must err on at least one. Both extensions carry over to the other lower bounds of Table 1, which come from the same counting argument (§3.5): each of them holds at every size $n \geq n_0(2^k)$, and each bound for targets that compute a query also holds for randomized targets, with the factor $1 - 2\varepsilon$; we do not restate these versions. The same comparison bounds how much better than guessing a circuit can do on random inputs, and this bound is exact.

Corollary 1.9 (Random inputs)**.** *Fix $k \geq 1$ and n. Color the edges of the complete graph on n vertices independently and uniformly with the colors in $2^{[k]}$, determine P_i, Q_i by the colors as in §1.3, and fix a vertex $\hbar$. Choose a color λ uniformly, a vertex v uniformly among those with $\chi(\hbar, v) = \lambda^c$ (any vertex if there is none), and a fair bit; if the bit is one, apply the fold of λ. Then every relational circuit with g composition gates answers $G_k(\hbar, v)$ correctly with probability at most $\frac{1}{2} + g/2^k + \delta$, where $\delta := n^3 4^k \exp(-(n-3)/4^k)$, and for each $g \leq 2^{k-1}$ some circuit with g composition gates answers correctly with probability at least $\frac{1}{2} + g/2^k - \delta$.*

1.5 Related work

Relation algebra and its representations. The calculus of relations predates first-order logic as a formal system: as De Morgan's logic of relations and as the logic of relatives of Peirce and Schröder, it was developed in the nineteenth century [DM64, Pei70, Sch95], decades before Hilbert's lectures and Hilbert and Ackermann's textbook isolated first-order logic [ES13, HA28, Moo88]. Its exact relation to first-order logic emerged gradually. Löwenheim published Korselt's observation that the calculus cannot express every first-order sentence [Löw15], Tarski axiomatized it [Tar41], and Tarski and Givant showed that it defines exactly the binary queries expressible with three variable names [TG87]; see also Maddux [Mad06]. Pratt, Maddux, Burris and Legris, and Brady survey this history [Pra92, Mad91, BL21, Bra00].

With N colors, the color-determined relations on our reference structure form the finite relation algebra $E_{N+1}(1, 2, 3)$. Jipsen, Maddux, and Tuza gave random and affine-plane representations of this algebra [JMT95]. Their affine-plane coloring, which assigns at least two directions to every color, already gives every pair of distinct vertices two witnesses for every pair of colors when the plane has order at least four; Appendix A verifies this. The probabilistic construction in §3.1 supplies such a coloring separately at every sufficiently large domain size, which our size-specific bounds need. We use these representations as reference structures. The additional ingredient for the lower bound is a bound on how many folds a single operation can notice, uniformly over all values on the reference structure; the existence of the finite representations alone does not provide an operation bound.

Succinctness and lower-bound methods. The classical FO^3-to-CoR translation has a single-exponential size bound, whereas the reverse translation from terms is linear; Appendix D gives the size analysis used here. In prior work, an exponential lower bound was established for positive CoR using two-edge witness queries, whose witness is constrained through both edges (x, z) and (z, y), and extended to the existential fragment Σ_1CoR [Nak22, §6]. That section also treats relational circuit size and gives a superpolynomial separation over a single binary relation symbol. Thus sharing was already covered for restricted target languages; the succinctness question for full CoR remained open. Our bounds allow arbitrary complement, converse, and shared intermediate results together. Our main family uses growing binary signatures, so the single-relation result and our main theorem address different restrictions. The conditional evidence of Nakamura and Uezato for full CoR [NU27, Corollary 40] comes from the 3-OV hypothesis.

Formula-size games and techniques derived from them give succinctness gaps for temporal, finite-variable, and modal logics [AI03, GS05, FvdHIK13, HV19]; §1.2 shows that, in full CoR, a term with two compositions already distinguishes Nakamura's structures. Ehrenfeucht–Fraïssé, pebble, and multi-structural games [Fra54, Ehr61, Imm82, Lib04, Imm81, FLRV21, CFI+24] capture quantifier depth, number of variables, and number of quantifiers, but G_k needs only one quantifier and three variables, and its DNF nests no compositions: the gap lies in the number of compositions alone. Succinctness gaps between equally expressive logics have also been obtained by automaton arguments, for instance between FO^2 and unary temporal logic [EVW02]. In communication complexity, rank and discrepancy bounds [MS82, KN97] and Karchmer–Wigderson games [KW90, Juk12] are insensitive to complement, but they measure a single matrix, and free Boolean operations defeat such measures: set disjointness has rank 2^k, yet its 1-entries form the complement of a union of k rectangles. Composition under complement alternates existential and universal quantification over witnesses, much as the communication polynomial hierarchy alternates nondeterminism and co-nondeterminism [BFS86]; there, lower bounds for explicit functions are open even against Arthur–Merlin protocols [GPW18]. We know of no transfer between the two settings. Dawar, Grohe, Kreutzer, and Schweikardt showed that the normal forms provided by Gaifman's locality theorem and by the Łoś–Tarski preservation theorem can require non-elementary size [DGKS07]; our result concerns, in the same spirit, the size cost of Tarski's correspondence between FO^3 and CoR. Our proof exploits the finite algebra of color-determined relations to bound how many folds each distinct operation of a circuit can notice, so it covers circuits with shared gates and unrestricted complement. For the variable-count comparison, Grohe and Schweikardt prove a polynomial translation from three-variable to two-variable sentences on pure finite linear orders [GS05, Theorem 4.3]. On linear orders they also show that four variables are exponentially more succinct than three: a fourth variable permits a divide-and-conquer recursion for distances, and three-variable formulas for distance 2^m need size $2^{\Omega(m)}$ [GS05]. A relational circuit achieves such a recursion by reuse, expressing distance 2^m with $m + 1$ compositions, so on linear orders circuits are exponentially more succinct than FO^3. Our lower bounds hold for circuits, so the gap they establish is not a matter of reuse. Vinall-Smeeth's variable-saving separations have at least three target variables and use higher-arity relations [VS24, Theorem 2]. Our sentence separation concerns three versus two variables on general finite structures with growing binary signatures. The linear FO^2-to-CoR translation is recorded in prior work [Nak22]; Proposition 1.5 shows an exponential cost in the other direction, even for FO^2-definable equations.

Matrices and query languages. In databases, Codd's relational algebra [Cod70, Cod72] made operations on whole relations the standard way to evaluate logical queries, and the calculus of relations is the binary core of navigational graph query languages [FGL+15, HPV+21, HVGW22]. A three-variable formula translates in linear size into relational algebra with at most three attributes per intermediate relation (§2), whereas CoR keeps every intermediate relation binary: a composition is a join followed by the projection of the shared attribute. Theorem 1.1 thus measures the cost of eliminating the third attribute. Binary intermediate relations are what fast matrix multiplication needs, and Boolean matrix products are a current tool for evaluating join–project queries [DHK20, Hu24, AHS25] and first-order graph properties [Wil14, GIKW17, BFK19]. Our bounds count such whole-relation operations; classical scalar-gate lower bounds for matrix products concern a different computation model [JS82, Juk12].

MATLANG combines matrix products, transpose, vector and diagonal operations, and entrywise functions [BGVW19, GMR+21]. For semiring-annotated relations [GKT07], Brijder, Gyssens, and Van den Bussche connect binary-output three-attribute relational algebra with matrix expressions [BGV22]. Their earlier manuscript asks whether exponential arity-elimination overhead is necessary [BGV19, p. 9]. The semiring version $\mathcal{G}_k$ of G_k gives such a lower bound over each fixed nontrivial commutative idempotent semiring of finite join breadth, even with arbitrary kernels in the typed circuit model of §6. That section specifies the permitted operations and separately treats positive nonidempotent computations.

1.6 Organization

Section 2 fixes semantics and resource measures. Section 3 builds the reference structure, proves that unnoticed folds leave a circuit's values as predicted and that each composition notices at most two folds, and states the test–cost theorem. Section 4 supplies the tests for G_k and proves Theorem 1.1 together with the other bounds for relational circuits: equations, fixed sizes and bounded error, circuits without converse, domain size, random inputs, and closure. Section 5 transfers the bound to quantifiers, determines the minimum term size, and proves Theorem 1.4 and Proposition 1.5. Section 6 extends the analysis of noticed folds to matrix languages, and Section 7 collects open questions. The appendices give explicit two-witness colorings, the term-size charging argument and the sharpness of the fold tests, the directed construction for circuits without converse, and the classical single-exponential translation.

2 Preliminaries

Structures and formulas. A signature is a finite set of binary relation symbols, without function or individual constant symbols. A structure $\mathcal{M}$ has a nonempty domain $\mathrm{dom}(\mathcal{M})$ and interprets each symbol R by a binary relation $R^{\mathcal{M}}$. Lower bounds require correctness only on finite structures; the explicit translations used for upper bounds hold on all nonempty structures. The logics FO^2 and FO^3 use only the variable names x, y and x, y, z, respectively, allowing repeated binding of names. Equality is available. The free-variable set of φ is $\mathrm{FV}(\varphi)$. If it is contained in $\{x, y\}$, write $[\![\varphi]\!]_{\mathcal{M}} \subseteq \mathrm{dom}(\mathcal{M})^2$ for its assignment relation. In particular, a sentence denotes the full or empty relation. All syntax uses unary negation and quantification and binary conjunction and disjunction; displayed iterated operations are abbreviations. The size $|\varphi|$ counts nodes after abbreviations are expanded. Thus $|G_k| = 8k$. Each relation-atom occurrence has unit cost, regardless of the spelling

of its relation name. This is a syntax-node measure over growing signatures, rather than a bit-length measure for encoded symbol names.

Evaluation and three-attribute intermediates. There is a direct $O(|\varphi|n^3)$ evaluation algorithm for an FO^3 formula on an n-element domain: store the truth table of each subformula over all assignments to x, y, z. Boolean operations act entrywise. For an existential quantifier, first take the disjunction along its coordinate and then repeat the resulting table along that coordinate. Each step takes $O(n^3)$ time, and universal quantification uses conjunction instead. In particular, G_k can be evaluated in $O(kn^3)$ time.

The same construction gives a linear-size translation into set relational algebra with at most three attributes per intermediate relation, using the unary domain relation as a constant. Lift every subformula to the common schema $\{x, y, z\}$. For a reflexive atom $R(u, u)$, select the diagonal of R and project to the attribute u; equality atoms are obtained by selection from domain products. Rename ordinary binary atoms as needed, and add missing domain coordinates by Cartesian product. Negation takes the difference from the full three-attribute domain relation. Existential quantification projects away its attribute and then restores that attribute by a Cartesian product with the domain. Each logical operation therefore uses a constant number of algebraic operations, and no intermediate has more than three attributes. For a binary assignment query, finally project to $\{x, y\}$, retaining either coordinate as a dummy if it is not free. This comparison concerns the permitted operations and schemas; our lower bounds below concern computation by the specified whole-relation or whole-matrix operations, rather than tuple-by-tuple evaluation time.

Terms and circuits. The CoR operations and constants have the meanings specified in the introduction. A term T is a syntax tree over binary input symbols, $\mathbf{0}, \mathbf{1}, \mathbf{id}$, union, intersection, complement, converse, and composition. Its value in $\mathcal{M}$ is $[\![T]\!]_{\mathcal{M}}$. Its size $|T|$ is its number of nodes, and $\#_{;}(T)$ counts composition occurrences. A relational circuit (DAG) C uses the same operations and one designated output, but may share intermediate results; $[\![g]\!]_{\mathcal{M}}$ denotes the value in $\mathcal{M}$ of a gate g, and $[\![C]\!]_{\mathcal{M}}$ that of the output. Write $|C|$ for its number of distinct nodes, including inputs, and $g_{;}(C)$ for its number of distinct composition gates. We subsequently say simply *relational circuit*. All operations other than composition may occur without any bound. For a tree,

$$|T| \geq 2\#_{;}(T) + 1,$$

since its leaves outnumber its binary nodes by one. For a term T, $\#_{\mathrm{rel}}(T)$ counts occurrences of input-relation symbols. For a logical formula φ, $\#_{\mathrm{rel}}(\varphi)$ counts occurrences of relation atoms, including reflexive atoms $R(x, x)$, and excludes equality atoms. Quantifier occurrences are counted by $\#_{\exists\forall}(\varphi)$.

Write $T \equiv_{\mathrm{fin}} \varphi$ when they define the same binary query on every finite structure over the input signature, and similarly for circuits and formulas. Unsubscripted equivalences used in explicit expansions hold on all nonempty structures. A size-specific circuit may depend on $n = |\operatorname{dom}(\mathcal{M})|$, but not on the input interpretations. No ordering, vertex predicates, named vertices, or arbitrary fixed $n \times n$ arrays are supplied for free. The only relational constants are the empty, full, and identity relations.

Randomized targets. A randomized circuit is an input-independent distribution of deterministic circuits over the permitted operations; its description may depend on n. Pointwise error at most ε

means that, for every input and each output position, the reported value is correct with probability at least $1-\varepsilon$. This need not be simultaneous correctness of the entire output relation. All randomized resource bounds below concern expectations over this distribution, whose support may contain incorrect or differently sized circuits. The same convention applies to randomized formulas and scalar computations.

Set and scalar notation. We use $[k] \coloneqq \{1,\dots,k\}$ and $2^{[k]}$ for its power set. For $A \subseteq [k]$, we write $A^c \coloneqq [k] \setminus A$. Empty relation intersections and unions mean $\mathbf{1}$ and $\mathbf{0}$; empty conjunctions and disjunctions mean true and false. Relation complement is always denoted by an overline and is relative to the full square of the current domain. Truth values are $\mathbb{B} \coloneqq \{0,1\}$, and brackets $[\mathcal{P}]$ mean the indicator of a statement $\mathcal{P}$. In the construction, V is the vertex set, $\hbar$ its hub, Λ its palette, and $N \coloneqq |\Lambda|$; from §3.2 on, $\Lambda = 2^{[k]}$ and $N = 2^k$. Colors are subsets of $[k]$; the letter λ denotes a color that is folded, and μ, ν denote arbitrary colors. A color-determined relation X has values $X[\mu]$ and $X[\mathbf{id}]$, and $X^{(\lambda)}$ denotes its prediction under the fold of λ (§3.2).

3 Folds, predictions, and the test–cost theorem

This section proves the machinery of the lower bounds in the order of the overview. We construct the reference structure (§3.1) and define color-determined relations, folds, and predictions (§3.2). We then show that a relational circuit computes the predicted values on every folded structure whose fold none of its compositions notices (§3.3), and that each composition notices at most two folds (§3.4). With the tests of §4.1, this already proves the composition bounds of Theorem 1.1. Section 3.5 states the test–cost theorem, a counting form of the same argument that also covers randomized circuits and the other targets of §1.4; Section 6 supplies its hypotheses for scalar-valued computations.

3.1 The reference structure

Fix a palette Λ of $N \geq 2$ colors and a vertex set V of size n. A coloring $\chi : \binom{V}{2} \to \Lambda$ assigns a color to every unordered pair of distinct vertices. We write $\chi(u,v)$ for $\chi(\{u,v\})$.

Lemma 3.1 (Two witnesses)**.** *For every integer*

$$n \geq n_0(N) \coloneqq \left\lceil 12N^2 \ln(2N) \right\rceil + 3,$$

there is a coloring such that, for all distinct $u, v \in V$ and all $\mu, \nu \in \Lambda$, at least two distinct vertices $z \in V \setminus \{u,v\}$ satisfy $\chi(u,z) = \mu$ and $\chi(z,v) = \nu$.

Proof. Color the edges independently and uniformly. For fixed distinct u, v, w and colors μ, ν, the probability that no suitable witness occurs outside $\{u,v,w\}$ is at most $\exp(-(n-3)/N^2)$: the candidate witnesses use disjoint pairs of edges. A union bound gives failure probability at most

$$n^3 N^2 \exp(-(n-3)/N^2).$$

At $n = n_0(N)$, the inequality $n_0(N) \leq 13N^3$ bounds this expression by $2197/(4096N) < 1$. It decreases for $n > 3N^2$, proving the assertion at every larger size. Hence some coloring has, for all distinct u, v, w and all colors μ, ν, a witness outside $\{u,v,w\}$; applying this with w equal to one witness for u, v yields a second. □

Fix such a coloring, and choose a vertex $\hbar \in V$, called the hub. The two witnesses of Lemma 3.1 play different roles. One witness outside the endpoints already makes every computed relation a union of color classes, possibly together with the diagonal (Lemma 3.2). The second ensures that two vertices other than $\hbar$ have, for every pair of colors, a witness that also avoids $\hbar$, so that recoloring the edges at $\hbar$ does not affect pairs away from it (§3.4). Lemma 3.1 also implies that every vertex sees every color, and that every vertex other than $\hbar$ sees every color on edges avoiding $\hbar$ (take $v = \hbar$). For each color μ, fix a vertex v_μ with $\chi(\hbar, v_\mu) = \mu$. No vertex, color, or color neighborhood is supplied as an additional input predicate.

Lemma 3.1 is a two-witness formulation of the colored extension construction of Jipsen, Maddux, and Tuza [JMT95], who gave both random and affine-plane colorings. Their affine-plane construction already supplies two witnesses when the plane has order at least four. We state the property in this form because our proof must keep every color pair available after excluding the hub. Proposition A.1 in Appendix A gives a self-contained two-witness affine-plane construction on fewer than $16N^2$ vertices.

3.2 Color-determined relations, folds, and predictions

For $\mu \in \Lambda$, let $e_\mu \coloneqq \{(u, v) \in V^2 : u \neq v,\ \chi(u, v) = \mu\}$ be the set of edges of color μ, in both orientations. We extend χ to loops by giving every loop the extra label $\mathbf{id}$, which is not a color. A relation $X \subseteq V^2$ is **color-determined** if $X(u, v)$ depends only on $\chi(u, v)$. For such a relation, write $X[\mu] \in \mathbb{B}$ for its common value at pairs (u, v) with $\chi(u, v) = \mu$, where $\mu \in \Lambda \cup \{\mathbf{id}\}$. These $N + 1$ values form a table, and

$$S_X \coloneqq \{\mu \in \Lambda : X[\mu] = 1\}$$

is the **off-diagonal support** of X. Thus X is the union of the classes e_μ with $\mu \in S_X$, together with the diagonal if $X[\mathbf{id}] = 1$. A reference structure interprets every input relation by a color-determined relation; we write $\mathcal{M}_0$ for it.

Lemma 3.2 (Closure of color-determined relations)**.** *The constants $\mathbf{0}, \mathbf{1}, \mathbf{id}$ are color-determined, and color-determined relations are closed under Boolean operations, converse, and composition. If X and Y are color-determined, then for every color $\mu \in \Lambda$,*

$$\begin{aligned} (X;Y)[\mu] &= (X[\mathbf{id}] \wedge Y[\mu]) \vee (X[\mu] \wedge Y[\mathbf{id}]) \vee (S_X \neq \varnothing \wedge S_Y \neq \varnothing), \\ (X;Y)[\mathbf{id}] &= (X[\mathbf{id}] \wedge Y[\mathbf{id}]) \vee (S_X \cap S_Y \neq \varnothing), \end{aligned} \tag{CT}$$

where a condition stands for its truth value.

Proof. Boolean operations act entry by entry, and converse preserves this property because χ is symmetric. For composition, consider a pair (u, v) and a potential witness z. An endpoint witness, $z \in \{u, v\}$, contributes the disjuncts of (CT) that involve $X[\mathbf{id}]$ or $Y[\mathbf{id}]$. An outside witness, $z \notin \{u, v\}$, contributes $X[\chi(u, z)] \wedge Y[\chi(z, v)]$. If $u \neq v$, then by Lemma 3.1 the color pairs $(\chi(u, z), \chi(z, v))$ of outside witnesses range over all of $\Lambda \times \Lambda$, so an outside witness exists exactly when S_X and S_Y are both nonempty. If $u = v$, both edges of an outside witness are the edge $\{u, z\}$, and every color occurs on an edge at u, so an outside witness exists exactly when $S_X \cap S_Y \neq \varnothing$. □

In particular, every value of a relational circuit on $\mathcal{M}_0$ is color-determined, and

$$e_\mu ; e_\nu = \begin{cases} \mathbf{1}, & \mu = \nu, \\ \overline{\mathbf{id}}, & \mu \neq \nu. \end{cases}$$

Thus the color-determined relations form the symmetric relation algebra $E_{N+1}(1,2,3)$ of Jipsen, Maddux, and Tuza [JMT95]; here $\overline{\mathbf{id}} = \bigcup_{\mu\in\Lambda} e_\mu$ is the diversity relation. The proof needs the displayed multiplication table, not automorphisms of the chosen finite graph.

Folds and predictions. From now on the palette is $\Lambda = 2^{[k]}$ for some $k \geq 1$, so that $N = 2^k$. The complement of a color μ is the color $\mu^c = [k] \setminus \mu$, and no color equals its complement. The fold of a color λ recolors every edge of color λ at the hub to λ^c and leaves all other edges unchanged: for $u \neq v$,

$$\chi_\lambda(u,v) \coloneqq \begin{cases} \lambda^c, & \hbar \in \{u,v\} \text{ and } \chi(u,v) = \lambda, \\ \chi(u,v), & \text{otherwise,} \end{cases}$$

and every loop keeps the label **id**. Only two properties of the fold matter below: it removes exactly the color λ from the hub, and it changes no edge away from the hub. Any target color other than λ would serve as well. For a color-determined relation X, its prediction $X^{(\lambda)}$ reads the same values on χ_λ:

$$X^{(\lambda)}(u,v) \coloneqq X[\chi_\lambda(u,v)] \qquad (u,v \in V).$$

Thus $X^{(\lambda)}$ agrees with X except on the recolored edges, where it takes the value $X[\lambda^c]$. Replacing every input relation by its prediction defines the folded structure $\mathcal{M}_\lambda$ on the same domain V. Predictions are defined only for color-determined relations, in particular for all values computed on $\mathcal{M}_0$. We do not assume that relations computed on $\mathcal{M}_\lambda$ are color-determined; in general, they are not.

3.3 Noticing, and unnoticed folds

Let λ be a color and $\mathbf{U} = (U_1, \ldots, U_r)$ a tuple of color-determined relations. An r-ary operation op notices the fold of λ at $\mathbf{U}$ if

$$\mathrm{op}(U_1^{(\lambda)}, \ldots, U_r^{(\lambda)}) \neq \mathrm{op}(\mathbf{U})^{(\lambda)}.$$

Both sides are determined by $\mathbf{U}$ and λ: noticing is a property of the operands on $\mathcal{M}_0$, not of any computation on $\mathcal{M}_\lambda$.

Proposition 3.3 (Commutation with prediction)**.** *For any color λ and color-determined relations X, Y, prediction commutes with Boolean operations and converse:*

$$(X \cup Y)^{(\lambda)} = X^{(\lambda)} \cup Y^{(\lambda)}, \quad (X \cap Y)^{(\lambda)} = X^{(\lambda)} \cap Y^{(\lambda)},$$
$$(\overline{X})^{(\lambda)} = \overline{X^{(\lambda)}}, \qquad (X^\smile)^{(\lambda)} = (X^{(\lambda)})^\smile.$$

The constants $\mathbf{0}, \mathbf{1}, \mathbf{id}$ are also fixed by prediction.

Proof. Boolean operations act pair by pair, so both sides apply the same operation to the entries at $\chi_\lambda(u,v)$. Converse commutes with prediction because $\chi_\lambda(u,v) = \chi_\lambda(v,u)$. The constants are fixed because a fold preserves the diagonal. □

Thus only composition can notice a fold. A composition gate of a relational circuit notices the fold of λ if the composition of its operand values on $\mathcal{M}_0$ notices it.

Lemma 3.4 (Unnoticed folds)**.** *Let C be a relational circuit and λ a color. If no composition gate of C notices the fold of λ, then on $\mathcal{M}_\lambda$ every gate of C computes the prediction of its value on $\mathcal{M}_0$. In particular, the outputs of C on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$ agree at every pair that the fold does not recolor.*

Proof. Evaluate C on $\mathcal{M}_0$. By Lemma 3.2, every gate value is color-determined, so its prediction is defined. We show by induction along a topological order of the gates that every gate computes on $\mathcal{M}_\lambda$ the prediction of its value on $\mathcal{M}_0$. For input relations this is the definition of $\mathcal{M}_\lambda$, and constants are their own predictions by Proposition 3.3. For Boolean and converse gates, the claim follows from the induction hypothesis and Proposition 3.3. If the operands of a composition gate have values X and Y on $\mathcal{M}_0$, then by induction they have values $X^{(\lambda)}$ and $Y^{(\lambda)}$ on $\mathcal{M}_\lambda$. The gate therefore computes $X^{(\lambda)};Y^{(\lambda)}$ on $\mathcal{M}_\lambda$, which equals $(X;Y)^{(\lambda)}$ because the gate does not notice the fold. Finally, every value U on $\mathcal{M}_0$ agrees with $U^{(\lambda)}$ at every pair that the fold does not recolor. □

3.4 Each composition notices at most two folds

Fix color-determined relations X, Y and a color λ, and write $J \coloneqq \Lambda \setminus \{\lambda\}$ for the set of colors at $\hbar$ after the fold of λ. Endpoint witnesses contribute the same value to $X^{(\lambda)};Y^{(\lambda)}$ and to $(X;Y)^{(\lambda)}$. At a pair (u,v) with $u \neq v$ and $\mu \coloneqq \chi_\lambda(u,v)$, they contribute $(X[\mathbf{id}] \wedge Y[\mu]) \vee (X[\mu] \wedge Y[\mathbf{id}])$ to $X^{(\lambda)};Y^{(\lambda)}$, because $X^{(\lambda)}$ and $Y^{(\lambda)}$ take the values $X[\mathbf{id}]$ and $Y[\mathbf{id}]$ on the diagonal and $X[\mu]$ and $Y[\mu]$ at (u,v), and the same value to $(X;Y)^{(\lambda)}(u,v) = (X;Y)[\mu]$ by (CT); at a diagonal pair, they contribute $X[\mathbf{id}] \wedge Y[\mathbf{id}]$ to both. The following table compares the outside witnesses. Its left column states when $X^{(\lambda)};Y^{(\lambda)}$ has an outside witness at the pair, and its right column states the last disjunct of (CT), which $(X;Y)^{(\lambda)}$ reads at the pair; here u and v range over vertices other than $\hbar$, and $S \neq \varnothing \neq S'$ means that both sets are nonempty.

Output pair	Outside witness in $X^{(\lambda)};Y^{(\lambda)}$	Last disjunct of (CT)
(u,v), $u \neq v$	$S_X \neq \varnothing \neq S_Y$	$S_X \neq \varnothing \neq S_Y$
(u,u)	$S_X \cap S_Y \neq \varnothing$	$S_X \cap S_Y \neq \varnothing$
$(\hbar,v)$	$S_X \cap J \neq \varnothing \neq S_Y$	$S_X \neq \varnothing \neq S_Y$
$(u,\hbar)$	$S_X \neq \varnothing \neq S_Y \cap J$	$S_X \neq \varnothing \neq S_Y$
$(\hbar,\hbar)$	$S_X \cap S_Y \cap J \neq \varnothing$	$S_X \cap S_Y \neq \varnothing$

For the left column, consider the outside witnesses z of a pair. For distinct $u, v \neq \hbar$, Lemma 3.1 gives, for every pair of colors, an outside witness other than $\hbar$, whose two edges keep their colors; the witness $\hbar$ contributes one more color pair, which is already realized. For (u,u) with $u \neq \hbar$, every color occurs on an edge at u that avoids $\hbar$. At a pair $(\hbar,v)$ of the **hub row**, the left edge $\{\hbar,z\}$ of every outside witness lies at the hub, so its color becomes $\chi_\lambda(\hbar,z) \in J$, while the right edge $\{z,v\}$ keeps its color; by Lemma 3.1 the realized color pairs are exactly $J \times \Lambda$. The pairs $(u,\hbar)$ of the **hub column** are symmetric, with $\Lambda \times J$. At $(\hbar,\hbar)$ both edges of an outside witness are the edge $\{\hbar,z\}$, whose color $\chi_\lambda(\hbar,z)$ ranges exactly over J.

Lemma 3.5 (When a composition notices a fold)**.** *Let X, Y be color-determined. The composition $X;Y$ notices the fold of λ only if*

$$S_X = \{\lambda\}, \qquad S_Y = \{\lambda\}, \qquad \text{or} \qquad S_X \cap S_Y = \{\lambda\}.$$

Hence it notices at most two folds, at most one if S_X or S_Y is $\varnothing$ or Λ, and none if both are.

Proof. By the table, the two sides can differ only in the hub row, in the hub column, or at $(\hbar, \hbar)$. In the hub row they can differ only if $S_X \neq \varnothing$ but $S_X \cap J = \varnothing$, that is, $S_X = \{\lambda\}$; in the hub column only if $S_Y = \{\lambda\}$; and at $(\hbar, \hbar)$ only if $S_X \cap S_Y = \{\lambda\}$. Each condition determines λ. If $S_X \cap S_Y \neq \varnothing$, each of the first two conditions implies the third, so at most one fold is noticed; otherwise the third condition never holds, and at most two are. If S_Y is $\varnothing$ or Λ, then $S_Y \neq \{\lambda\}$ because $N \geq 2$, and $S_X \cap S_Y$ is $\varnothing$ or S_X, so only the condition $S_X = \{\lambda\}$ remains; the case of S_X is symmetric. If both are $\varnothing$ or Λ, none of the conditions can hold. □

The bound two is attained. The composition $e_\mu; e_{\mu^c}$ has $S_X = \{\mu\}$ and $S_Y = \{\mu^c\}$, and notices the folds of both μ and μ^c. On the reference structure of §4.1, the relation e_μ is the relation H_μ of the DNF. Proposition B.1 computes G_k on the reference structure and all its folded structures with 2^{k-1} such compositions. By Lemma 3.5, each composition gate notices at most two folds. What a gate notices depends only on its operand values on $\mathcal{M}_0$, so a shared gate counts once, regardless of its fan-out. Hence at most $2g_;(C)$ folds are noticed by some gate of C, and Lemma 3.4 applies to all other folds. Nothing here depends on the query or on the interpretation of the input relations. If a query has different answers on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$ at a pair that the fold of λ does not recolor, for every color λ, then every relational circuit C defining it satisfies $2g_;(C) \geq 2^k$. Section 4 shows that G_k is such a query.

3.5 Tests and the test–cost theorem

For randomized circuits and for the other targets of §1.4, we use a counting form of this argument that refers only to a few abstract properties. For a color λ, the **test** for λ consists of the structures $\mathcal{M}_0$ and $\mathcal{M}_\lambda$ and the **output observation** $\rho_\lambda(U) \coloneqq U(\hbar, v_{\lambda^c})$, the entry of an output value U at the pair $(\hbar, v_{\lambda^c})$ of §3.1. Since $\lambda^c \neq \lambda$, the fold of λ does not recolor this pair, so

$$\rho_\lambda(U^{(\lambda)}) = \rho_\lambda(U) \tag{O}$$

for every reference output value U, as defined in (i) below.

The counting argument uses only three properties of this construction. We call data satisfying them a **test system**:

(i) Each type of value has a collection of **reference values**, which contains the input values on $\mathcal{M}_0$ and is closed under the permitted operations, and every reference value U has a prediction $U^{(\lambda)}$ for each color λ.

(ii) Every input value on $\mathcal{M}_\lambda$ is the prediction of the same input value on $\mathcal{M}_0$.

(iii) Every test satisfies (O), and the required answers at its observation on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$ differ.

For relational circuits, the reference values are the color-determined relations, and Lemma 3.2 supplies (i). The typed scalar computations in §6 will satisfy the same conditions; predictions carry the type of their argument, which we suppress in the notation. For a gate g of a deterministic circuit C, let op_g be its operation and $\mathbf{U}_g$ the tuple of its operand values on $\mathcal{M}_0$. As in §3.3, g notices the fold of λ if $\mathrm{op}_g(\mathbf{U}_g^{(\lambda)}) \neq \mathrm{op}_g(\mathbf{U}_g)^{(\lambda)}$, where predictions act componentwise on tuples. Define

$$D_g \coloneqq \{\lambda \in \Lambda : g \text{ notices the fold of } \lambda\},$$
$$\mathrm{cost}(C) \coloneqq \sum_{g \text{ distinct}} |D_g|.$$

For input nodes put $D_g \coloneqq \varnothing$; constant gates are nullary operations. The definition of D_g compares the entire output of the gate, and it uses predicted operands computed from $\mathcal{M}_0$, not operands recomputed on $\mathcal{M}_\lambda$.

We call $\operatorname{cost}(C)$ the **cost** of C. It counts the pairs of a distinct gate and a fold that the gate notices. This integer depends on the reference structure and the tests, which remain fixed, but is defined even for an incorrect circuit.

Theorem 3.6 (Test–cost theorem)**.** *Fix a test system. A deterministic circuit C correct at both observations of every test satisfies*

$$2^k \le \operatorname{cost}(C).$$

If C_ω is chosen from an input-independent distribution of circuits, and each observation in each test is incorrect with probability at most $\varepsilon < 1/2$, then

$$(1 - 2\varepsilon)2^k \le \mathbb{E}_\omega \operatorname{cost}(C_\omega).$$

Proof. Fix a deterministic circuit, not assumed correct, and put $D_C \coloneqq \bigcup_g D_g$. For $\lambda \notin D_C$, no gate notices the fold of λ, so the induction in the proof of Lemma 3.4, with (ii) for the input values, gives

$$[\![C]\!]_{\mathcal{M}_\lambda} = ([\![C]\!]_{\mathcal{M}_0})^{(\lambda)}.$$

The equality concerns the whole output. By the observation condition (O), the two observations are equal, whereas their correct values differ. Hence at least one is wrong. If $\operatorname{err}_{\lambda,0}(C), \operatorname{err}_{\lambda,1}(C)$ are the corresponding zero-one error indicators, then

$$\operatorname{cost}(C) + \sum_{\lambda \in \Lambda} \big(\operatorname{err}_{\lambda,0}(C) + \operatorname{err}_{\lambda,1}(C)\big) \ge 2^k,$$

because $|D_C| \le \operatorname{cost}(C)$. Zero error proves the first assertion. Take expectations using the same random circuit on the two structures of each test. Each error indicator has expectation at most ε, which proves the second assertion. No independence among the tests or their errors is required. □

This is the counting argument for all subsequent lower bounds on the reference structure. Each of them bounds $\operatorname{cost}(C)$ from above by a resource count of the target, such as $2g_;(C)$ in Lemma 3.7 below, and therefore also holds in two further forms, which we do not restate. First, Lemma 3.1 gives the reference structure on every domain of size $n \ge n_0(2^k)$, and the folds keep the domain, so each bound holds for targets that need to be correct only on n-element inputs and may be chosen for n. Second, by the randomized assertion of Theorem 3.6, each bound for targets that compute a query also holds for an input-independent distribution of targets with pointwise error at most $\varepsilon < 1/2$, after the resource count is replaced by its expectation and the bound is multiplied by $1 - 2\varepsilon$; for scalar outputs, an error is a failure to return the exact required value.

For ordinary CoR, call a composition gate whose operand values on $\mathcal{M}_0$ are X, Y **one-sided** if S_X or S_Y is $\varnothing$ or Λ. Let $m_1(C)$ count these gates and $m_2(C)$ the other composition gates, counting shared gates once.

Lemma 3.7 (Cost of compositions)**.** *Every relational circuit C satisfies*

$$\operatorname{cost}(C) \le m_1(C) + 2m_2(C) \le 2g_;(C).$$

Proof. By Lemma 3.5, a one-sided composition notices at most one fold and any other composition at most two. Other gates notice nothing. Summing over distinct gates proves the inequalities. □

The projections $X;\mathbf{1}$ and $\mathbf{1};X$ are one-sided. If a syntactic count is preferred, classify only these projections as one-sided and put every remaining composition in m_2; Lemma 3.7 still holds.

4 Lower bounds for relational circuits

The construction of §3 is independent of the query. We now interpret the input relations of G_k on it; the read edge of each test identifies the witness color that the fold removes.

4.1 The tests for G_k

On a color μ, interpret P_i as 1 precisely when $i \in \mu$, and Q_i as 1 precisely when $i \notin \mu$. Put both values zero on every loop. More explicitly,

$$P_i^{\mathcal{M}_0} := \bigcup_{\mu\in\Lambda:\, i\in\mu} e_\mu, \qquad Q_i^{\mathcal{M}_0} := \bigcup_{\mu\in\Lambda:\, i\notin\mu} e_\mu.$$

For each color λ, use the read edge $(\hbar, v_{\lambda^c})$, of color λ^c (§3.1), and the fold of λ (§3.2). Since $\lambda \neq \lambda^c$, the fold does not recolor this read edge.

Lemma 4.1 (The tests for G_k)**.** *For every color λ,*

$$[\![G_k]\!]_{\mathcal{M}_0}(\hbar, v_{\lambda^c}) = 1, \qquad [\![G_k]\!]_{\mathcal{M}_\lambda}(\hbar, v_{\lambda^c}) = 0.$$

Moreover, the sentence $G_k^\forall = \forall x\,\forall y\,(x = y \vee G_k(x,y))$ of §1.4 is true in $\mathcal{M}_0$ and false in every $\mathcal{M}_\lambda$. Thus G_k and $G_k^\forall$ each supply 2^k tests.

Proof. For non-loop edges with colors μ, ν, the body of G_k holds exactly when $\nu = \mu^c$. A loop, whose P_i, Q_i values are all zero, cannot be a witness edge. Hence, on $\mathcal{M}_0$ and on every $\mathcal{M}_\lambda$, G_k holds at a pair (u, v) of distinct vertices exactly when u has an edge whose color is the complement of the color of (u, v). Against the read color λ^c, the sole accepted witness color is λ. That color occurs at $\hbar$ in $\mathcal{M}_0$ and does not occur there after the fold. In $\mathcal{M}_0$, every vertex has edges of every color, so G_k holds at every pair of distinct vertices, and $x = y$ handles the diagonal; in $\mathcal{M}_\lambda$, G_k fails at the pair $(\hbar, v_{\lambda^c})$ of distinct vertices. □

4.2 Consequences

Theorem 1.1 (Exponential lower bounds for CoR)**.** *For every $k \geq 1$, every relational circuit $C \equiv_{\text{fin}} G_k$ has at least 2^{k-1} composition gates, and every* CoR *term $T \equiv_{\text{fin}} G_k$ has at least 2^{k-1} compositions.*

Proof. Let $C \equiv_{\text{fin}} G_k$, and take the reference structure at any size $n \geq n_0(2^k)$. For each color λ, the answers of G_k at $(\hbar, v_{\lambda^c})$ on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$ differ by Lemma 4.1, and the fold of λ does not recolor this pair. By Lemma 3.4, some composition gate of C notices the fold of λ. By Lemma 3.5, each composition gate notices at most two folds, so $2g_;(C) \geq 2^k$. A term is a circuit without sharing, so the same bound holds for its number of compositions. □

Together with the expansion of G_k in §1.1, these lower bounds show that the minimum circuit size for G_k is $\Theta(2^k)$; Theorem 5.3 in §5.2 determines the minimum term size.

The lower bound does not rest on the difficulty of evaluating G_k. As its proof shows, it holds even for circuits that are correct only on the reference structure and its folded structures, on which every non-loop edge has a color from the palette. A read edge there accepts exactly the witness edges of the complementary color, so a trie of the colors present in each row evaluates G_k on all pairs of these structures in time $O(kn^2)$, linear in the size of the input.

Equations. The same tests bound the equational form of Tarski's correspondence.

Corollary 1.3 (CoR equations)**.** *If a* CoR *equation $T = S$ holds on exactly those finite structures that satisfy $G_k^\forall$, then every relational circuit with gates computing T and S has at least $2^{k-1} - 1$ composition gates.*

Proof. Let C have gates computing T and S, and add gates for

$$\mathrm{Eq} \coloneqq \overline{\mathbf{1}; ((T \cap \overline{S}) \cup (\overline{T} \cap S)); \mathbf{1}},$$

which is the full relation when $T = S$ and empty otherwise. This adds two compositions, both one-sided. By Lemma 4.1, $G_k^\forall$ is true in $\mathcal{M}_0$ and false in every $\mathcal{M}_\lambda$, so Eq is full on $\mathcal{M}_0$ and empty on every $\mathcal{M}_\lambda$. Both constants are their own predictions, so these are 2^k tests, and Theorem 3.6 with Lemma 3.7 gives $2^k \le 2g_;(C) + 2$. □

Fixed sizes and bounded error. The reference structure and its folded structures share one domain, which may have any size $n \ge n_0(2^k)$, and the randomized assertion of Theorem 3.6 handles bounded error.

Corollary 1.8 (Fixed sizes and bounded error)**.** *Fix $k \ge 1$, $n \ge n_0(2^k)$, and $0 \le \varepsilon < 1/2$. Every input-independent distribution of relational circuits computing G_k on all n-element inputs with pointwise error at most ε satisfies*

$$\mathbb{E}\, g_;(C) \ge (1 - 2\varepsilon)2^{k-1}.$$

In particular, each exact circuit at each such size needs at least 2^{k-1} composition gates, even when chosen specifically for n.

Proof. At the prescribed size n, Lemma 3.1 gives the reference structure, and Lemma 4.1 gives all 2^k tests on that domain. Apply the expected-cost assertion of Theorem 3.6 and $\mathrm{cost}(C) \le 2g_;(C)$. Setting $\varepsilon = 0$ includes every exact size-specific circuit. □

Circuits without converse. Without converse, a directed version of the reference structure lets each composition notice at most one fold, and the bound becomes exact.

Corollary 4.2 (Without converse)**.** *For every $k \ge 1$, every relational circuit without converse that defines G_k on finite structures has at least 2^k composition gates. This bound is attained by the DNF.*

Appendix C proves the corollary with a directed reference coloring. For full CoR, Proposition B.1 gives a circuit with 2^{k-1} compositions that is correct on the reference structure and all its folded structures. Thus these tests attain the coefficient two of Lemma 3.5 and cannot give more than 2^{k-1}. The optimum for G_k on arbitrary inputs remains between 2^{k-1} and 2^k.

4.3 Domain size

On arbitrary inputs, the **profile** $\pi(x,z) \in \{0,1\}^{2k}$ of a pair lists the values of $P_1,\dots,P_k,Q_1,\dots,Q_k$ at (x,z); on the reference structure, the profile of an edge is determined by its color. By the definition of G_k, the read edge (x,y) accepts the witness edge (x,z) exactly when, for every i, the profile $\pi(x,y)$ has P_i and $\pi(x,z)$ has Q_i, or $\pi(x,y)$ has Q_i and $\pi(x,z)$ has P_i. Hence $G_k(x,y)$ depends only on $\pi(x,y)$ and the set $\Pi_x \coloneqq \{\pi(x,z) : z \in V\}$ of profiles in row x, which has at most $|V|$ elements.

Proposition 4.3 (Small domains)**.** *For all $k,n \geq 1$ there is a converse-free relational circuit with $4kn$ composition gates, all of them row projections $X;\mathbf{1}$, and $O(k^2n^2)$ other gates, that computes G_k on every structure with at most n elements.*

Proof. A relation is **row-constant** if its value at (x,y) does not depend on y. Every row projection $S;\mathbf{1}$ is row-constant, and it holds in row x exactly when some pair (x,z) lies in S; Boolean combinations of row-constant relations are row-constant. For $1 \leq b \leq 2k$, let E_b be the input relation giving bit b of the profile, and put $E_b^1 \coloneqq E_b$ and $E_b^0 \coloneqq \overline{E_b}$.

In a structure with at most n elements, call a word $w \in \{0,1\}^\ell$ **live** in row x if some profile in Π_x begins with w. Live words of one length begin distinct profiles, so there are at most n of each length, and the empty word is live because $\pi(x,x) \in \Pi_x$. The circuit stores the live words of length ℓ in n slots: row-constant relations $L_{\ell,r}$ and $W_{\ell,r,b}$, for $r \leq n$ and $b \leq \ell$, state that slot r is occupied in row x and that its word has bit b. Initially only slot 1 is occupied, by the empty word. For each slot r and each $\sigma \in \{0,1\}$, the relation

$$S_{\ell,r,\sigma} = L_{\ell,r} \cap \bigcap_{b \leq \ell} \big((E_b \cap W_{\ell,r,b}) \cup (\overline{E_b} \cap \overline{W_{\ell,r,b}})\big) \cap E^\sigma_{\ell+1}$$

holds at (x,z) exactly when slot r is occupied in row x and $\pi(x,z)$ begins with its word followed by σ. The row projection $S_{\ell,r,\sigma};\mathbf{1}$ therefore tells, in every row at once, whether this extension is live. The at most n live extensions are placed into the slots of length $\ell+1$ in the order of their pairs (r,σ). This uses only Boolean operations on the $2n$ answers and the stored words, $O(kn^2)$ gates per length, since the rank of each live extension is a count of earlier live ones. After length $2k$ the occupied slots hold exactly the profiles in Π_x, and

$$G_k = \bigcup_{r \leq n} \Big(L_{2k,r} \cap \bigcap_{i=1}^{k} \big((P_i \cap W_{2k,r,k+i}) \cup (Q_i \cap W_{2k,r,i})\big)\Big),$$

because the intersection over i says that the read edge (x,y) accepts the witness edges whose profile is in slot r. The circuit makes $2n$ row projections at each of the $2k$ lengths, $4kn$ in all, and uses no converse. □

Proposition 1.2 (Domain size)**.** *For some absolute constant $C > 0$, for every $k \geq 1$ and $n \geq Ck4^k$, every relational circuit computing G_k on all n-element structures has at least 2^{k-1} composition gates.*

Proof. By Corollary 1.8 with $\varepsilon = 0$, the lower bound holds at every domain size $n \geq n_0(2^k)$. Since

$$n_0(2^k) = \lceil 12 \cdot 4^k (k+1) \ln 2 \rceil + 3 \leq 18k4^k$$

for all $k \geq 1$, we may take $C = 18$. □

4.4 Random inputs

Corollary 1.9 (Random inputs)**.** *Fix $k \geq 1$ and n. Color the edges of the complete graph on n vertices independently and uniformly with the colors in $2^{[k]}$, determine P_i, Q_i by the colors as in §1.3, and fix a vertex $\hbar$. Choose a color λ uniformly, a vertex v uniformly among those with $\chi(\hbar, v) = \lambda^c$ (any vertex if there is none), and a fair bit; if the bit is one, apply the fold of λ. Then every relational circuit with g composition gates answers $G_k(\hbar, v)$ correctly with probability at most $\frac{1}{2} + g/2^k + \delta$, where $\delta := n^3 4^k \exp(-(n-3)/4^k)$, and for each $g \leq 2^{k-1}$ some circuit with g composition gates answers correctly with probability at least $\frac{1}{2} + g/2^k - \delta$.*

Proof. Let $\mathcal{W}$ be the event that the coloring has the two-witness property. The proof of Lemma 3.1, with $N = 2^k$, gives $\Pr[\neg\mathcal{W}] \leq \delta$. Fix a coloring in $\mathcal{W}$. Then $\hbar$ has edges of every color, so v is uniform among the vertices with $\chi(\hbar, v) = \lambda^c$. For each color λ, let $\mathrm{err}_{\lambda,0}, \mathrm{err}_{\lambda,1}$ indicate an error of C at $(\hbar, v_{\lambda^c})$ on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$. The error-count inequality in the proof of Theorem 3.6, with the tests of Lemma 4.1 and Lemma 3.7, gives

$$\sum_{\lambda} (\mathrm{err}_{\lambda,0} + \mathrm{err}_{\lambda,1}) \geq 2^k - \mathrm{cost}(C) \geq 2^k - 2g$$

for every choice of the vertices v_μ of §3.1, hence also in expectation when each v_μ is uniform among the vertices with $\chi(\hbar, v_\mu) = \mu$. Given the coloring, the probability of an error is the average of $\mathrm{err}_{\lambda,b}$ over uniform λ and b, so it is at least $(2^k - 2g)/2^{k+1} = \frac{1}{2} - g/2^k$. Averaging over the colorings proves the upper bound.

For the lower bound, choose g of the pairs $\{\lambda, \lambda^c\}$ and keep, in the circuit of Proposition B.1, only the terms for these pairs. This circuit has g composition gates. On a coloring in $\mathcal{W}$, it is correct on both structures of the $2g$ tests whose colors lie in the chosen pairs, and for every other color it answers zero at the read edge, which is correct when the bit is one. Its success probability is therefore at least $(1 - \delta)(\frac{1}{2} + g/2^k) \geq \frac{1}{2} + g/2^k - \delta$. □

4.5 Reflexive–transitive closure

Proposition 1.7 (Closure does not help)**.** *Let C be a relational circuit that may also use $X \mapsto X^*$, and let $g_*(C)$ count its distinct closure gates. If $C \equiv_{\mathrm{fin}} G_k$, then $2g_;(C) + g_*(C) \geq 2^k$; in particular, at least 2^{k-1} gates of C are compositions or closures.*

Proof. On the reference structure, (CT) gives $X; X = \mathbf{1}$ whenever the off-diagonal support S_X is nonempty. Hence $X^* = \mathbf{1}$ if $S_X \neq \varnothing$ and $X^* = \mathbf{id}$ otherwise; in particular, closure preserves color-determined relations, as condition (i) of a test system requires. Let λ be a color, and put $J := \Lambda \setminus \{\lambda\}$. If $S_X = \varnothing$, then $X^{(\lambda)} = X$, and both $(X^{(\lambda)})^*$ and $(X^*)^{(\lambda)}$ equal $\mathbf{id}$. If S_X contains a color in J, the table of §3.4 gives $X^{(\lambda)}; X^{(\lambda)} = \mathbf{1}$, so $(X^{(\lambda)})^* = \mathbf{1} = (X^*)^{(\lambda)}$. Thus a closure gate notices the fold of λ only when $S_X = \{\lambda\}$, so it notices at most one fold. Theorem 3.6 and Lemma 3.7 give $2^k \leq m_1(C) + 2m_2(C) + g_*(C) \leq 2g_;(C) + g_*(C)$ with the tests of Lemma 4.1. □

5 Two-variable logic and tree size

The formula G_k is FO^2-definable, but its displayed linear-size form uses three variable names. This section quantifies the difference, proves the term-size bound of Theorem 5.3, and separates CoR

from FO^2 (§5.4).

5.1 A translation that preserves the relevant resources

For a formula φ using only x, y, view its value as a binary relation of assignments, even if one or both variables are absent from $\mathrm{FV}(\varphi)$. Thus a sentence denotes $\mathbf{1}$ or $\mathbf{0}$. The standard linear translation τ into CoR is as follows:

φ	$\tau(\varphi)$	φ	$\tau(\varphi)$
$R(x,y)$	R	$R(y,x)$	$R^\smile$
$R(x,x)$	$(R \cap \mathbf{id}); \mathbf{1}$	$R(y,y)$	$\mathbf{1}; (R \cap \mathbf{id})$
$x = y$ or $y = x$	$\mathbf{id}$	$x = x$ or $y = y$	$\mathbf{1}$
$\exists x\, \psi$	$\mathbf{1}; \tau(\psi)$	$\exists y\, \psi$	$\tau(\psi); \mathbf{1}$

Conjunction, disjunction, and negation translate to intersection, union, and complement. Universal quantifiers use $\forall u\, \psi \equiv \neg\exists u\, \neg\psi$, for $u \in \{x, y\}$. Induction on formulas proves correctness on all nonempty structures. The linear size bound is recorded in [Nak22, Remark 3.22].

Translation preserves the relation-atom count $\#_{\mathrm{rel}}$ exactly. On the reference structure, the off-diagonal support of $R \cap \mathbf{id}$ is empty. Hence the extra composition used for a reflexive atom notices no fold. Every quantifier contributes a one-sided composition, which notices at most one fold. These observations give the following resource bound.

Lemma 5.1 (Quantifier cost)**.** *On the reference structure, every* FO^2 *formula* φ *satisfies*

$$\mathrm{cost}(\tau(\varphi)) \leq \#_{\exists\forall}(\varphi).$$

This is a statement about cost, not about the syntactic composition count: the latter also counts the compositions introduced by reflexive atoms, which notice no fold.

Combining Lemma 5.1 with Theorem 3.6 and the tests of Lemma 4.1 gives

$$\varphi \equiv_{\mathrm{fin}} G_k \Longrightarrow \#_{\exists\forall}(\varphi) \geq 2^k.$$

The FO^2 expansion obtained from the DNF of §1.1 uses exactly 2^k quantifiers, so the optimum quantifier count for G_k is exactly 2^k. No restriction to existential target formulas is made.

5.2 Counting input occurrences in trees

The test–cost theorem also yields a stronger bound for syntax trees. A **singleton selector** for a color λ is a Boolean formula in the tables of the input relations, with constants allowed, whose support on the entire palette is exactly $\{\lambda\}$.

Lemma 5.2 (Disjoint input-occurrence charging)**.** *Fix the reference structure and its folds. Suppose every singleton selector requires at least* $d > 0$ *input-relation occurrences. Then every* CoR *term* T*, whether correct or not, satisfies*

$$\mathrm{cost}(T) \leq \#_{\mathrm{rel}}(T)/d.$$

Appendix B.1 proves the lemma by assigning disjoint sets of input occurrences to composition nodes. The disjointness uses the tree structure of a term.

For the input relations of §4.1, a Boolean formula in their tables that contains neither P_i nor Q_i takes the same value on a color and on its i-th-bit flip. Hence a singleton selector contains P_i or Q_i for every i, so $d = k$, and this is exact, since H_λ selects λ. Theorem 3.6 and Lemma 5.2 give $\#_{\mathrm{rel}}(T) \geq k2^k$ for every CoR term $T \equiv_{\mathrm{fin}} G_k$. A syntax tree whose nodes have at most two children has at least $2L - 1$ nodes if it has L leaves, and $L \geq \#_{\mathrm{rel}}(T)$. Therefore

$$|T| \geq 2k2^k - 1 \qquad (T \equiv_{\mathrm{fin}} G_k),$$

which is the lower bound of the following theorem.

Theorem 5.3 (Term size)**.** *For every $k \geq 1$, every* CoR *term $T \equiv_{\mathrm{fin}} G_k$ has at least $2k2^k - 1$ nodes, and G_k is equivalent to a* CoR *term with $(2k + 6)2^k - 5$ nodes.*

Proof. The lower bound is the inequality above. Factoring the conditions on the read edge into a binary decision tree gives a term with $(2k + 6)2^k - 5$ nodes that is equivalent to G_k on all nonempty structures (Appendix B.2). □

5.3 Sentences and the variable hierarchy

We now prove the sentence separation stated in the introduction, for the sentence $G_k^\forall = \forall x \, \forall y \, (x = y \vee G_k(x, y))$.

Theorem 1.4 (Three-variable versus two-variable sentences)**.** *For each $k \geq 1$, the sentence $G_k^\forall$ has size $O(k)$ and quantifier depth three over $2k$ binary relations. It is* FO^2*-definable, but every equivalent* FO^2 *sentence over finite structures has at least 2^k quantifiers and $k2^k$ relation-atom occurrences. Its minimum* FO^2 *tree size is $\Theta(k2^k)$.*

Proof. The source has size $8k + 4$ and quantifier depth three. By Lemma 4.1, it is true in $\mathcal{M}_0$ and false in every $\mathcal{M}_\lambda$. A sentence is represented by a constant binary relation, and either constant is its own prediction. Thus these structures give 2^k tests for the sentence itself. For any equivalent FO^2 sentence φ, apply Theorem 3.6 to $\tau(\varphi)$. Lemma 5.1 gives $\#_{\exists\forall}(\varphi) \geq 2^k$, and Lemma 5.2, together with exact preservation of $\#_{\mathrm{rel}}$, gives $\#_{\mathrm{rel}}(\varphi) \geq k2^k$. Conversely, substitute the FO^2 expansion of G_k into the sentence. It has $2^k + 2$ quantifiers and $O(k2^k)$ nodes. This proves FO^2-definability and the matching order of tree size. □

The quantifier optimum for this sentence is therefore in $[2^k, 2^k + 2]$; the exact 2^k optimum proved earlier concerns the binary query.

This complements the variable-saving separations of Vinall-Smeeth [VS24, Theorem 2], whose stated range starts at three variables on the target side. Our result concerns the remaining comparison of three and two variables, over growing binary signatures. It does not claim a lower bound for separating a single prescribed pair of structures as a function of their size. On pure finite linear orders, in contrast, Grohe and Schweikardt [GS05, Theorem 4.3] give a polynomial translation from three-variable sentences to two-variable sentences. The qualifications "pure orders" and "sentences" are essential to that comparison.

5.4 Separating CoR from FO^2

The linear translation above goes from FO^2 to CoR. On the reference structure, a composition notices at most two folds and a quantifier at most one, so tests on the reference structure bound CoR circuits and FO^2 formulas alike, up to a factor of two. To separate the two formalisms, we use a structure on which a quantifier still notices at most one fold, but a single composition notices all of them.

Use input symbols $R, P_1, \dots, P_k$, and put

$$D_i^1 \coloneqq \mathbf{id} \cap P_i, \qquad D_i^0 \coloneqq \mathbf{id} \cap \overline{P_i}, \qquad T_k \coloneqq R; \bigcap_{i=1}^{k} \left[(D_i^0; \mathbf{1}; D_i^0) \cup (D_i^1; \mathbf{1}; D_i^1) \right].$$

The complement in D_i^0 is taken before intersection with the identity; it is not the complement of D_i^1. Expanding these abbreviations gives an $O(k)$-size tree. For a vertex v, call $\pi(v) \coloneqq \{i \in [k] : P_i(v, v)\}$ its **loop pattern**. Then $T_k(x, y)$ holds exactly when some R-successor z of x satisfies $\pi(z) = \pi(y)$, that is, when

$$\varphi_k(x, y) \coloneqq \exists z \left(R(x, z) \wedge \bigwedge_{i=1}^{k} (P_i(z, z) \leftrightarrow P_i(y, y)) \right)$$

holds. Here $\alpha \leftrightarrow \gamma$ abbreviates $(\alpha \wedge \gamma) \vee (\neg\alpha \wedge \neg\gamma)$.

The structures. Let $V \coloneqq 2^{[k]} \times \mathbb{Z}_{11}$ and $\Gamma \coloneqq \{1, 2, 3, 4, 7, 8\} \subseteq \mathbb{Z}_{11}$. The structure $\mathcal{W}_0$ on V has the loops $P_i((A, s), (A, s))$ for $i \in A$, no other P_i-entries, and $R((A, s), (B, s'))$ exactly when $s' - s \in \Gamma$; thus $\pi(A, s) = A$, and R has no loops. The **atomic type** of a pair (u, v) of distinct vertices is $(R(u, v), R(v, u), \pi(u), \pi(v))$, and that of a loop (u, u) is $\pi(u)$; it determines the values of all atomic formulas at the pair. As $s' - s$ ranges over $\mathbb{Z}_{11} \setminus \{0\}$, the truth values of $s' - s \in \Gamma$ and $s - s' \in \Gamma$ take each of the four combinations at least twice: $(1, 0)$ at $s' - s \in \{1, 2\}$, $(0, 1)$ at $s' - s \in \{9, 10\}$, $(1, 1)$ at $s' - s \in \{3, 4, 7, 8\}$, and $(0, 0)$ at $s' - s \in \{5, 6\}$. Hence every vertex u is the first entry of at least two pairs of each atomic type $(r, r', \pi(u), B)$, and the second entry of at least two pairs of each atomic type $(r, r', B, \pi(u))$.

Fix the hub $\hbar \coloneqq (\varnothing, 0)$. For $A \subseteq [k]$, the fold of A yields the structure $\mathcal{W}_A$ obtained from $\mathcal{W}_0$ by removing from R the edges $(\hbar, v)$ with $v \neq \hbar$ and $\pi(v) = A$. It changes the atomic types of these pairs and of their converses, and of no other pair. Afterwards, no pair with first entry $\hbar$ has a type $(1, r', \varnothing, A)$, and no pair with second entry $\hbar$ has a type $(r, 1, A, \varnothing)$; all other types still occur at $\hbar$.

Type tables and predictions. A relation $X \subseteq V^2$ has a **type table** if $X(u, v)$ depends only on the atomic type of (u, v) in $\mathcal{W}_0$; let S_X be the set of atomic types at which X holds. Every atomic type of $\mathcal{W}_A$ occurs in $\mathcal{W}_0$, so the prediction $X^{(A)}$ can read the same table at the atomic types of $\mathcal{W}_A$. As in §3.3, an operation op notices the fold of A at operand values $\mathbf{U}$ if $\mathrm{op}(\mathbf{U}^{(A)}) \neq \mathrm{op}(\mathbf{U})^{(A)}$, and Boolean operations never notice a fold.

View every FO^2 formula as defining a binary relation, as in §5; its quantifiers then act as the projections $\exists y\, \psi \equiv \psi; \mathbf{1}$ and $\exists x\, \psi \equiv \mathbf{1}; \psi$. By induction on the formula, every relation that an FO^2 formula defines on $\mathcal{W}_0$ has a type table: if X has one, then $(X; \mathbf{1})(u, v)$ holds exactly when S_X contains the atomic type of some pair with first entry u, and these types are determined by $\pi(u)$; symmetrically for $\mathbf{1}; X$.

Lemma 5.4 (Projections notice at most one fold)**.** *If X has a type table, then $X;\mathbf{1}$ and $\mathbf{1};X$ each notice the fold of at most one $A \subseteq [k]$.*

Proof. At a pair (u, v), the prediction of $X;\mathbf{1}$ holds exactly when S_X contains the atomic type in $\mathcal{W}_0$ of some pair with first entry u, and $X^{(A)};\mathbf{1}$ holds exactly when S_X contains the atomic type in $\mathcal{W}_A$ of some such pair. If $u \neq \hbar$, the fold changes at most one pair with first entry u, namely $(u, \hbar)$; since every type occurs at least twice among these pairs in $\mathcal{W}_0$, the two sets of types coincide. If $u = \hbar$, the fold removes exactly the types $(1, r', \varnothing, A)$ and adds none. Hence $X;\mathbf{1}$ notices the fold of A only if the types in S_X of pairs with first entry $\hbar$ are all of the form $(1, r', \varnothing, A)$, and there is at least one; this determines A. The argument for $\mathbf{1};X$ uses second entries. □

Proposition 1.5 (CoR versus FO^2)**.** *For every $k \geq 1$, the* CoR *equation $\mathbf{id} \cup T_k = \mathbf{1}$ is FO^2-definable, but every equivalent FO^2 sentence over finite structures has at least 2^k quantifiers. Every FO^2 formula defining T_k also has at least 2^k quantifiers, and this bound is attained.*

Proof. For the upper bounds, write $\vartheta_A(u) \coloneqq \bigwedge_{i \in A} P_i(u, u) \wedge \bigwedge_{i \in A^c} \neg P_i(u, u)$ for a variable u. The FO^2 formula

$$\bigvee_{A \subseteq [k]} \left[\vartheta_A(y) \wedge \exists y \left(R(x, y) \wedge \vartheta_A(y)\right)\right]$$

defines T_k with exactly 2^k quantifiers; the inner y is rebound only in the second conjunct. The equation $\mathbf{id} \cup T_k = \mathbf{1}$ says that for all x and all $y \neq x$, some R-successor of x has the loop pattern of y, and so does the FO^2 sentence

$$\forall x \bigwedge_{A \subseteq [k]} \left[\exists y \left(\neg x = y \wedge \vartheta_A(y)\right) \to \exists y \left(R(x, y) \wedge \vartheta_A(y)\right)\right].$$

For the lower bounds, every vertex of $\mathcal{W}_0$ has an R-successor with each loop pattern, so T_k is the full relation on $\mathcal{W}_0$ and the equation holds there. On $\mathcal{W}_A$, the hub has no R-successor with loop pattern A, so $T_k(\hbar, v)$ fails for every $v \neq \hbar$ with $\pi(v) = A$, and the equation fails. Let ψ be an FO^2 formula defining T_k, or an FO^2 sentence equivalent to the equation. Its value on $\mathcal{W}_0$ is the full relation, whose prediction on $\mathcal{W}_A$ is again full, but its value on $\mathcal{W}_A$ is not. If no quantifier of ψ noticed the fold of A, then, as in the proof of Lemma 3.4, every subformula would take its predicted value on $\mathcal{W}_A$, by induction on the formula: atomic types determine atomic formulas, and Boolean operations notice nothing. Hence some quantifier notices the fold of each A, and by Lemma 5.4 each quantifier notices at most one, so ψ has at least 2^k quantifiers. □

6 Scalar aggregation and matrix query languages

Relational composition aggregates two edge values over a common index. The calculation of §3.4 extends to scalar aggregation without any order or monotonicity assumption on the operation combining those values. This gives arity-elimination lower bounds for matrix query languages.

6.1 Finite join breadth and arbitrary kernels

Let $(S, \vee, 0_S)$ be a join-semilattice with least element 0_S. Its join breadth β is the least nonnegative integer such that every finite join equals the join of at most β of its summands. The empty join is

0_S. We assume $\beta < \infty$. A totally ordered set with maximum as join has breadth one. The space $\{0,1\}^b$ with coordinatewise disjunction has breadth b.

Let V be the common finite index set of the input matrices, with $n \coloneqq |V|$. For any function $\kappa : S \times S \to S$, called a kernel, define

$$(X ;_\kappa Y)(u,v) \coloneqq \bigvee_{z \in V} \kappa(X(u,z), Y(z,v)).$$

Kernels may differ between gates. They need not be associative, monotone, or zero-preserving. We also allow arbitrary uniform entrywise functions, transpose, uniform scalar constants, and a diagonal mask with two fixed values, one on and one off the diagonal. "Uniform" means the same function is used at all positions of a given type; it has no access to vertex names. The inputs to the query are square $n \times n$ arrays. No order, vertex identifiers, arbitrary input vectors, or vertex-dependent constant arrays are supplied in addition to them.

Each circuit has a fixed directed acyclic wiring. Every entrywise function, kernel, and scalar constant is chosen before seeing the input; it may depend on n and k in a size-specific circuit. An entrywise function receives the scalar operands at its position, and a kernel receives only the two scalar arguments displayed above. In particular, a kernel does not also receive $X(u,v)$, the position names u, v, z, or an encoding of the whole input unless such a value has been computed and supplied through its two operands by permitted operations. Data-dependent values and tests are allowed, but there is no primitive for data-dependent rewiring or indexed access to arbitrary entries. These conventions specify the target model used in Theorem 1.6.

Reference arrays are color-determined in the sense of §3.2, now with values $X[\mathbf{id}], X[\mu] \in S$. Predictions read these values on the folded coloring χ_λ. Entrywise functions and transpose preserve this property and notice no fold. For color-determined arrays X, Y and $J \subseteq \Lambda$, define

$$\kappa^{\mathrm{L}}_J \coloneqq \bigvee_{\mu \in J,\, \nu \in \Lambda} \kappa(X[\mu], Y[\nu]), \qquad \kappa^{\mathrm{R}}_J \coloneqq \bigvee_{\mu \in J,\, \nu \in \Lambda} \kappa(X[\nu], Y[\mu]),$$
$$\kappa^{\mathrm{D}}_J \coloneqq \bigvee_{\mu \in J} \kappa(X[\mu], Y[\mu]), \qquad \kappa_* \coloneqq \kappa^{\mathrm{L}}_\Lambda = \kappa^{\mathrm{R}}_\Lambda.$$

The reference product $Z \coloneqq X ;_\kappa Y$ has the table

$$Z[\mathbf{id}] = \kappa(X[\mathbf{id}], Y[\mathbf{id}]) \vee \kappa^{\mathrm{D}}_\Lambda,$$
$$Z[\mu] = \kappa(X[\mathbf{id}], Y[\mu]) \vee \kappa(X[\mu], Y[\mathbf{id}]) \vee \kappa_*.$$

The calculation of §3.4 remains valid, with endpoint contributions formed by κ and, away from the hub, outside-witness contributions κ_* off the diagonal and $\kappa^{\mathrm{D}}_\Lambda$ on it. For the fold of λ, the outside-witness contributions in the hub row, in the hub column, and at $(\hbar, \hbar)$ become κ^{L}_J, κ^{R}_J, and κ^{D}_J with $J = \Lambda \setminus \{\lambda\}$, whereas the prediction of $X ;_\kappa Y$ reads them with $J = \Lambda$. This follows from the same color pairs of outside witnesses; idempotence of join makes multiplicities irrelevant.

Lemma 6.1 (Scalar local bounds)**.** *A scalar aggregation gate notices at most 3β folds. If one operand is constant across off-diagonal colors, it notices at most β; if both are, it notices none. For Boolean scalars with arbitrary kernels, the general bound improves to two.*

Proof. By the calculation above, the gate notices the fold of λ only if one of the joins κ^{L}_J, κ^{R}_J, and κ^{D}_J with $J = \Lambda \setminus \{\lambda\}$ differs from its value at $J = \Lambda$. Choose at most β summands of $\kappa^{\mathrm{L}}_\Lambda$ whose join

is $\kappa^{\mathrm{L}}_{\Lambda}$, and let K^{L} be the set of their row colors, the row color of $\kappa(X[\mu], Y[\nu])$ being μ. If $\lambda \notin K^{\mathrm{L}}$, all chosen summands also occur in κ^{L}_{J}, so $\kappa^{\mathrm{L}}_{\Lambda} \le \kappa^{\mathrm{L}}_{J} \le \kappa^{\mathrm{L}}_{\Lambda}$. Define K^{R} in the same way from the column colors of summands of $\kappa^{\mathrm{R}}_{\Lambda}$, and K^{D} from the colors of summands of $\kappa^{\mathrm{D}}_{\Lambda}$. Hence the gate notices the fold of λ only if λ lies in $K^{\mathrm{L}} \cup K^{\mathrm{R}} \cup K^{\mathrm{D}}$, which has at most 3β colors.

If $Y[\mu]$ does not depend on μ, then $\kappa^{\mathrm{L}}_{J} = \kappa^{\mathrm{D}}_{J}$ and $\kappa^{\mathrm{R}}_{J} = \kappa_*$ for every nonempty J, so only K^{L} is needed; the case of X is symmetric. If both tables are constant, all three joins are unchanged. For Boolean scalars, if $\kappa^{\mathrm{D}}_{\Lambda} = 1$, a single color μ with $\kappa(X[\mu], Y[\mu]) = 1$ supplies a summand of value one to all three joins whenever $\mu \in J$, so only the fold of μ can be noticed. Otherwise the diagonal join is always zero, and one color for each of the other two joins suffices. □

These calculations show closure of color-determined arrays and bound the number of folds that each gate notices, as Theorem 3.6 requires; §6.3 supplies the query tests. Uniform entrywise functions, transpose, and the permitted constants notice no fold.

6.2 Typed matrix operations

We use the dimension symbols n and 1. Intermediate arrays have shapes $n \times n$, $n \times 1$, $1 \times n$, or 1×1; these are matrices, column vectors, row vectors, and scalars, respectively. An admissible contraction of an $a \times n$ and an $n \times c$ array is

$$Z(i,j) = \bigvee_{z=1}^{n} \kappa(X(i,z), Y(z,j)), \qquad a, c \in \{1, n\}.$$

Products contracting an index of size one are just applications of the kernel and are not data-dimension contractions. We allow entrywise functions, transpose, scalar and vector broadcasting, diagonal extraction and formation, uniform constant vectors, and the diagonal mask. All of these are uniform and do not aggregate a data index. This is at least as expressive as the corresponding MATLANG fragment with semiring multiplication as kernel; standard MATLANG operations are described in [BGVW19, BGV22].

On the reference structure, every vector is constant over its data index and every scalar is a single value. Encode a column vector $\mathbf{c}$ as the square array $(u, v) \mapsto \mathbf{c}(u)$, a row vector $\mathbf{r}$ as $(u, v) \mapsto \mathbf{r}(v)$, and a scalar as a constant square array. Reference closure follows from the calculation in §6.1. The encoded reference vector has the same value on every color and the diagonal, so it is its own prediction. Diagonal formation and extraction and all broadcasting operations notice no fold. Matrix–vector and vector–matrix contractions notice at most β folds by Lemma 6.1. A vector inner product notices none, since both operand values on the reference structure are constant and are their own predictions. Products over a size-one index notice none either.

Let $\#_{\mathrm{mv}}(C)$ count matrix–vector and vector–matrix contractions, and let $\#_{\mathrm{mm}}(C)$ count square matrix–matrix contractions, always counting shared gates once. The local bounds give the following typed resource estimate.

Lemma 6.2 (Cost of typed products)**.** *Every typed matrix circuit C in the model above satisfies*

$$\mathrm{cost}(C) \le \beta(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)).$$

For Boolean scalars and arbitrary kernels, the bound improves to $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C)$.

Proof. Sum the bounds of Lemma 6.1 over the distinct product gates. The other permitted operations notice no fold, as established above. □

The gate-by-gate accounting can be summarized as follows.

Operation	Folds noticed, at most
Uniform entrywise function, transpose, permitted constant	0
Diagonal formation or extraction, broadcasting	0
Product over an index of size one	0
Vector inner product	0
Matrix–vector or vector–matrix contraction	β
Square matrix–matrix contraction	3β

The cost counts noticed folds; an operation that notices none may still take time to evaluate. Inner products contribute zero to the cost.

6.3 Arity elimination for a weighted query

Let $(S, \vee, \otimes, 0_S, 1_S)$ be a nontrivial commutative idempotent semiring of finite join breadth β. Thus addition $\vee$ is an idempotent commutative monoid, multiplication $\otimes$ is a commutative monoid, multiplication distributes over addition, and 0_S annihilates multiplication. For the common input index set V, define

$$\mathcal{G}_k(x, y) \coloneqq \bigvee_{z \in V} \bigotimes_{i=1}^{k} \left[(P_i(x, y) \otimes Q_i(x, z)) \vee (Q_i(x, y) \otimes P_i(x, z)) \right].$$

An annotated relation assigns a scalar in S to each tuple. In positive annotated relational algebra, union is pointwise addition, natural join multiplies annotations, and projection sums over removed attributes. Renaming changes attribute names. We write ARA(3) for this language restricted to intermediate relations with at most three attributes. The weighted query $\mathcal{G}_k$ has an ARA(3) expression of size $O(k)$: form the two ternary joins in each clause, take their union, join the clauses, and project out z.

Theorem 1.6 (Exponential arity-elimination overhead)**.** *For every such semiring and $k \geq 1$, every typed matrix circuit C computing $\mathcal{G}_k$ on all finite inputs satisfies*

$$\beta(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)) \geq 2^k,$$

and so has at least $2^k/(3\beta)$ aggregation gates. This holds with arbitrary entrywise functions, each applied in the same way at every entry, and arbitrary kernels in the model of §6. Conversely, some typed matrix circuit with 2^k matrix–vector products computes $\mathcal{G}_k$ on all inputs.

For Boolean scalars, the same argument gives the stronger inequality $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C) \geq 2^k$, which §6.4 uses.

Proof. Use the $0_S, 1_S$ encodings of the Boolean G_k tests. Nontriviality makes these values distinct, and idempotence makes the value of $\mathcal{G}_k$ exactly its Boolean truth value under this encoding. Thus there are 2^k tests. Apply Theorem 3.6 and Lemma 6.2, or its Boolean refinement; the bound $2^k/(3\beta)$ on the number of aggregation gates follows since $\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C) \le 3(\#_{\mathrm{mv}}(C) + \#_{\mathrm{mm}}(C))$.

The upper bound uses exactly 2^k matrix–vector contractions. Define H_A by replacing the intersections in its earlier definition by pointwise products $\otimes$. For each A, contract H_{A^c} with the constant 1_S column vector, broadcast the result along rows, multiply entrywise by H_A, and join the outputs over A. Distributivity and commutativity prove that this computes $\mathcal{G}_k$ on arbitrary inputs. All H_A and H_{A^c} can be shared with $O(2^k)$ total nodes. □

Examples of breadth-one semirings include max–min on $[0,1]$, $(\mathbb{R} \cup \{-\infty\}, \max, +, -\infty, 0)$, and $(\mathbb{R} \cup \{+\infty\}, \min, +, +\infty, 0)$. For each, at least $2^k/3$ data-dimension products are necessary, even in the enlarged kernel model.

6.4 Positive computations without idempotence

A separate transfer handles positive arithmetic, for which join breadth is not available. Let $(S, +, \otimes, 0_S, 1_S)$ be a nontrivial commutative semiring satisfying

$$r\,1_S \neq 0_S \qquad \text{for every integer } r \ge 1, \tag{NV}$$

where $r\,1_S$ is the sum of r copies of 1_S. Restrict the target to positive entrywise addition and multiplication, semiring matrix products, transpose, broadcasting, diagonal operations, and constants in $\mathbb{N}1_S := \{r\,1_S : r \ge 0\}$. Arbitrary entrywise decoding and negative constants are not allowed here. Replace joins in $\mathcal{G}_k$ by sums to define its positive semiring version.

On Boolean-encoded inputs, every intermediate belongs to $\mathbb{N}1_S$. The map sending 0_S to zero and every other element of this subsemiring to one is a homomorphism to the Boolean semiring. Indeed, sums and products of positive multiples of 1_S are again nonzero by the nonvanishing condition (NV). Applying this map to every gate of a target circuit C gives a Boolean matrix circuit with exactly the same typed product counts. Its output is G_k. The Boolean lower bound therefore gives $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C) \ge 2^k$. Here the cost is applied to the Booleanized computation; no argument by predictions is claimed for the original scalar circuit. The 2^k-product expansion still works by distributivity. This includes positive computations over $\mathbb{N}$ and nonnegative reals.

The restriction on the operations matters: with nonidempotent addition and arbitrary entrywise decoding, a single aggregation along each row can encode how many edges (x, z) have each profile of §4.3, and these counts determine G_k. The arbitrary-kernel result of Theorem 1.6 avoids this because finite join breadth limits how many colors can support a single aggregation.

7 Open questions

Our bounds settle exponential FO^3-to-CoR compilation overhead over growing binary signatures. Several quantitative questions remain.

A fixed signature. Does a superpolynomial gap persist for full CoR over a fixed finite signature, or even one binary relation symbol? The encoding of [Nak22, §6] gives a superpolynomial separation

for the existential fragment over a single symbol. Conditional evidence points the same way for full CoR: under the 3-OV hypothesis, the argument of [NU27, Corollary 40] excludes polynomial-time translations already for existential positive FO^3 sentences over one binary relation symbol [NU27, Theorem 39]. Extending the unconditional conclusion to full CoR requires controlling unrestricted complement on the encoded inputs. Our present formulas use one or two binary symbols per coordinate.

Domain size. Propositions 1.2 and 4.3 place the onset of the blowup between domain sizes $2^k/(8k)$ and $O(k4^k)$. Our reference structures cannot be much smaller: for fixed distinct u, v, the 4^k pairs of colors have disjoint sets of at least two witnesses each, so a reference structure has at least $2 \cdot 4^k + 2$ vertices. The upper bound, on the other hand, uses only row projections. Locating the onset may require a different reference structure or circuits whose compositions combine different rows.

The exact composition count for G_k. On arbitrary finite inputs the optimum lies between 2^{k-1} and 2^k; without converse it is exactly 2^k. Proposition B.1 attains 2^{k-1} on the reference structure and all its folded structures. Consequently these tests alone cannot improve the full-CoR lower bound. Determining the optimum on arbitrary inputs calls for either a more efficient general circuit or tests exposing an additional obstruction.

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A Explicit two-witness colorings

This appendix gives a deterministic version of Lemma 3.1 on selected domain sizes. The affine-plane construction follows the direction-partition method of Jipsen, Maddux, and Tuza [JMT95]. Their assignment of at least two directions to every color already guarantees two witnesses at plane order at least four; we include the verification because this is the property used here.

Proposition A.1 (Explicit two-witness colorings)**.** *For every palette size $N \geq 2$, a graph with the two-witness property of Lemma 3.1 can be constructed deterministically on fewer than $16N^2$ vertices.*

Proof. Let s be the smallest power of two with $s \geq 2N - 1$, and let $\mathbb{F}_s$ be the field of s elements. Use vertex set $V := \mathbb{F}_s^2$. There are $s + 1$ directions of affine lines. Partition these directions into sets Dir_μ, $\mu \in \Lambda$, each of size at least two. Color distinct points u, v by the unique μ containing the direction of their joining line. Since $N \geq 2$, we have $s \geq 4$ and $|V| = s^2 < 16N^2$.

Fix distinct u, v and requested colors μ, ν. If $\mu = \nu = \chi(u, v)$, all $s - 2 \geq 2$ other points of the line uv are witnesses. Otherwise, choose a direction from Dir_μ at u and a direction from Dir_ν at v, neither equal to the direction of uv, and choose the two directions to be distinct. Their two lines intersect at a unique witness outside $\{u, v\}$. There are at least two choices: if the colors differ, their direction sets are disjoint and at most one set loses a direction to uv; if they agree but differ from $\chi(u, v)$, two directions from their common set can be taken in either order. Distinct direction pairs give distinct intersection points, since a point outside $\{u, v\}$ determines both directions uniquely. Thus the coloring has the two-witness property. □

For $N = 2^k$, one can choose $s = 2N$, giving 4^{k+1} vertices. This construction supplies selected domain sizes, rather than every large size; induced subgraphs need not retain the two-witness property.

B Tree bounds and limits of the method

B.1 The input-occurrence charging lemma

This subsection proves the tree refinement stated in §5.2.

Lemma 5.2 (Disjoint input-occurrence charging)**.** *Fix the reference structure and its folds. Suppose every singleton selector requires at least $d > 0$ input-relation occurrences. Then every* CoR *term T, whether correct or not, satisfies*

$$\mathrm{cost}(T) \leq \#_{\mathrm{rel}}(T)/d.$$

Proof. For a subterm whose value on the reference structure is X, let $S_X \subseteq \Lambda$ be the off-diagonal support of X. Boolean operations act on these sets in the usual way and converse fixes them. By the identities (CT), a composition $X;Y$ with both supports nonempty has off-diagonal support Λ. Call such a node **active**. If either support is empty, the off-diagonal table of the product is zero or one of the two operand tables, according to the diagonal bits. Such an inactive node notices no fold: every outside-witness contribution in the local calculation vanishes.

Construct a Boolean formula for the off-diagonal table of any subterm by replacing active compositions by the constant true, inactive compositions by zero or the relevant operand table, and converse by its argument. This operation never duplicates an input occurrence. For each active node $g = (X;Y)$, apply this pruning to its two operands. Let $\mathcal{L}_g$ be the set of original input occurrences that remain in the two resulting table formulas. These are the occurrences that *survive the pruning*, not all input occurrences below g. This distinction prevents repeated charging at successive active nodes. The sets $\mathcal{L}_g$ are pairwise disjoint: an active descendant is replaced by a constant when an ancestor's operands are pruned, and incomparable nodes have disjoint subtrees.

By Lemma 3.5, a composition notices the fold of a color λ only if

$$S_X = \{\lambda\}, \qquad S_Y = \{\lambda\}, \qquad \text{or} \qquad S_X \cap S_Y = \{\lambda\}.$$

Let E_g be the set of colors satisfying at least one of these conditions. Thus $|D_g| \le |E_g|$. If $S_X \cap S_Y$ is nonempty, E_g has size at most one. Its unique color, if present, is selected by one operand table or by their conjunction. The selector assumption therefore gives $|\mathcal{L}_g| \ge d|E_g|$. If the two supports are disjoint, only singleton operands contribute to E_g. Each contributing operand has at least d input occurrences, and the operands are disjoint trees. The same inequality follows. Summing over active nodes proves

$$\mathrm{cost}(T) = \sum_g |D_g| \le \sum_g |E_g| \le \frac{1}{d} \sum_g |\mathcal{L}_g| \le \#_{\mathrm{rel}}(T)/d.$$

Diagonal bits may affect which operand table an inactive node retains, but do not create additional input occurrences in this argument. □

B.2 Factored trees for G_k

A complete binary decision tree successively chooses P_i or Q_i for the read edge. At leaf A place $H_{A^c};\mathbf{1}$. A branch at coordinate i with child terms U_1, U_0 has the form

$$(P_i \cap U_1) \cup (Q_i \cap U_0).$$

Distributing the branches recovers the DNF for G_k on arbitrary inputs; no assumption that exactly one of P_i, Q_i holds at each edge is used. Each of the 2^k leaves costs $2k+1$ nodes, and each of the $2^k - 1$ internal branches adds five. The total is $(2k+6)2^k - 5$. The number of relation-input occurrences is $k2^k + 2(2^k - 1)$.

B.3 Why the requirements are crossed

For comparison, the superficially similar non-crossed formula collapses:

$$\exists z \bigwedge_{i=1}^{k} [(P_i(x,y) \wedge P_i(x,z)) \vee (Q_i(x,y) \wedge Q_i(x,z))] \equiv \bigwedge_{i=1}^{k} (P_i(x,y) \vee Q_i(x,y)).$$

The forward implication is immediate; for the reverse implication take $z = y$. In G_k, the two edges must meet different requirements at every coordinate, so the choice $z = y$ no longer works.

B.4 Sharpness of the fold tests

Proposition B.1 (Sharpness on the folded structures)**.** *For every $k \geq 1$, the minimum number of composition gates in a relational circuit that computes G_k on the reference structure and on all its folded structures is 2^{k-1}.*

Proof. For each unordered complementary pair $\{A, A^c\}$ of colors, choose one representative A and compute, once,

$$X_A \coloneqq H_A; H_{A^c}.$$

Use the shared result in

$$\bigcup_{\{A,A^c\}} \left[(H_{A^c} \cap X_A) \cup (H_A \cap X_A^{\smile})\right].$$

This circuit has exactly 2^{k-1} compositions. We check that it computes G_k on $\mathcal{M}_0$ and on every $\mathcal{M}_\lambda$. Let J be the set of colors at the hub, so that $J = \Lambda$ on $\mathcal{M}_0$ and $J = \Lambda \setminus \{\lambda\}$ on $\mathcal{M}_\lambda$. On each of these structures, H_A is the set of edges of color A. The relation X_A is true on all pairs of distinct non-hub vertices, has zero diagonal, and has hub-row and hub-column values $[A \in J]$ and $[A^c \in J]$, respectively. The masks select the complementary read colors. At a non-hub vertex every color still occurs, so G_k is true on every off-diagonal read edge there. At the hub it is true precisely when the complementary color lies in J. These are exactly the displayed circuit's values. The lower bound is Theorem 1.1, whose proof uses only these structures. □

C Circuits without converse

We prove the sharper bound for circuits without converse with a directed version of the reference structure.

Corollary 4.2 (Without converse)**.** *For every $k \geq 1$, every relational circuit without converse that defines G_k on finite structures has at least 2^k composition gates. This bound is attained by the DNF.*

Proof. Color each ordered pair of distinct vertices independently from the palette $\Lambda = 2^{[k]}$; write $\chi(u, v)$ for its color, with no relation between $\chi(u, v)$ and $\chi(v, u)$. Require that for every u, v, now allowing $u = v$, and every $\mu, \nu \in \Lambda$, at least two vertices z outside $\{u, v\}$ realize

$$(\chi(u, z), \chi(z, v)) = (\mu, \nu).$$

Candidate witnesses use disjoint ordered arcs, even when $u = v$, so the proof of Lemma 3.1 applies with the same threshold $n_0(N)$ and at most n^3N^2 bad events. Interpret P_i and Q_i on each arc by its color as in §4.1, with all loop entries zero.

Reference relations have a diagonal value $X[\mathbf{id}]$ and one value $X[\mu]$ per arc color μ; let S_X be the set of colors on which X holds. These tables are closed under Boolean operations and composition: endpoint witnesses contribute as in (CT), and every pair, including a diagonal pair, has an outside witness exactly when S_X and S_Y are both nonempty. The fold of λ now recolors only the arcs of

color λ that leave $\hbar$, to λ^c, and leaves all arcs entering $\hbar$ unchanged; predictions read the tables on this new coloring. Outside the row of $\hbar$, the composition of the predicted operands equals the prediction of the composition, since for every pair of colors some outside witness other than $\hbar$ uses two unchanged arcs. In the row of $\hbar$, including $(\hbar, \hbar)$, an outside witness exists exactly when $S_X \setminus \{\lambda\}$ and S_Y are both nonempty. Hence a composition notices the fold of λ only if $S_X = \{\lambda\}$, and so it notices at most one fold.

Choose v_μ so that the arc $(\hbar, v_\mu)$ has color μ, and read the output at $(\hbar, v_{\lambda^c})$. The query G_k inspects only arcs leaving x, so the first assertion of Lemma 4.1 holds with the same proof, reading the edges at a vertex as the arcs leaving it, and Theorem 3.6 gives $g_{;}(C) \geq 2^k$ for relational circuits without converse. The DNF of G_k uses no converse and attains this bound on arbitrary inputs. Converse is excluded because the color of an arc does not determine the color of its reverse arc; the coefficient-one argument consequently does not apply to full CoR. □

D The classical single-exponential translation

We include the size analysis for the classical expressive equivalence [TG87, Mad06]. The relevant point is that successive quantifier eliminations need not create a tower of exponentials: an already translated subformula is treated as one leaf of the next Boolean expansion, without expanding its algebraic representation again.

Proposition D.1 (Classical translation)**.** *Every* FO^3 *formula* φ *over binary relations with at most two free variables has an equivalent* CoR *term of size* $2^{O(|\varphi|)}$ *on all nonempty structures. Here size counts syntax-tree nodes.*

Proof. We represent a formula with at most two free variables as a binary relation of assignments to an ordered pair containing those variables. An unused coordinate is a dummy coordinate: membership is independent of it. Thus a unary formula is represented by a relation constant along rows or columns, and a sentence by $\mathbf{0}$ or $\mathbf{1}$. If a representation is moved to another ordered pair containing the same free variables, it is unchanged or replaced by its converse. This also handles the placement of a unary free variable in either coordinate. Closed formulas need no adjustment.

An atom with two distinct variables is represented by its input relation, its converse, or $\mathbf{id}$, as appropriate. A reflexive atom $R(u, u)$, when u is the first coordinate, is represented by $(R \cap \mathbf{id}); \mathbf{1}$; when it is the second coordinate, use the converse. An equality $u = u$ is represented by $\mathbf{1}$. These translations have constant size. Boolean operations on formulas over a fixed pair become intersection, union, and complement.

Process quantified subformulas from the inside out. Consider $\exists w\, \psi$, and let u, v be the other two variable names, in that order. Its free variables are contained in $\{u, v\}$. In the body ψ, regard each atom and each maximal quantified subformula as one leaf of a Boolean formula, called its **shell**. Each leaf has at most two free variables: this is immediate for binary atoms, and a quantifier binds one of the three available names. The quantified leaves have already been translated. Crucially, the shell expansion treats each of their translations as an opaque term.

If the shell has r leaf occurrences, a truth-table DNF has at most 2^r clauses, each using at most r literals. Repeated or logically dependent leaves cause no problem; the propositional identity remains valid after substituting their actual interpretations. In each clause, assign its literals to three groups according to the free-variable pair: (u, v), (u, w), or (w, v). A literal with only one free variable is

assigned to any group containing it; a closed literal can be put in the first group. Use converse to orient each translated literal for its chosen pair, and use complement for a negative literal. Writing the conjunctions of the groups as $A(u, v)$, $B(u, w)$, and $C(w, v)$, the clause translates as

$$\exists w\,(A(u, v) \wedge B(u, w) \wedge C(w, v)) \quad \longmapsto \quad A \cap (B; C).$$

An empty group is represented by $\mathbf{1}$. This identity also covers vacuous quantification because domains are nonempty. Take the union of the translated clauses. Universal quantification uses $\forall w\,\psi \equiv \neg\exists w\,\neg\psi$, which introduces no subformula duplication. At the root, any remaining Boolean shell with at most two free variables is translated directly, after orienting its leaves to a common ordered pair. This proves correctness on all nonempty structures by induction.

For the size bound, cut the original syntax tree at quantifiers into the shells just used. Count each child quantified subformula as one leaf of its parent's shell. The sum of the shell sizes, including these boundary leaves, is $O(|\varphi|)$: each original node is counted once, with only one extra boundary occurrence per quantified subformula. A shell of size s creates at most $2^{O(s)}$ local nodes and at most $2^{O(s)}$ copies of each child translation. Along any path from an original leaf or a newly introduced local node to the final output, the resulting duplication factor is therefore at most

$$\prod_{\text{shells on the path}} 2^{O(s)} \leq 2^{O(|\varphi|)}.$$

There are $O(|\varphi|)$ original nodes and shell locations, and the local exponential creation factor is included in this same bound. Summing their contributions gives $|\varphi|\, 2^{O(|\varphi|)} = 2^{O(|\varphi|)}$ nodes. No step expands the internal Boolean structure of a translated child, which is what prevents an iterated-exponential recurrence. □